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Machine learning molecular dynamics: exploring MACE

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a.y. 2025/2026

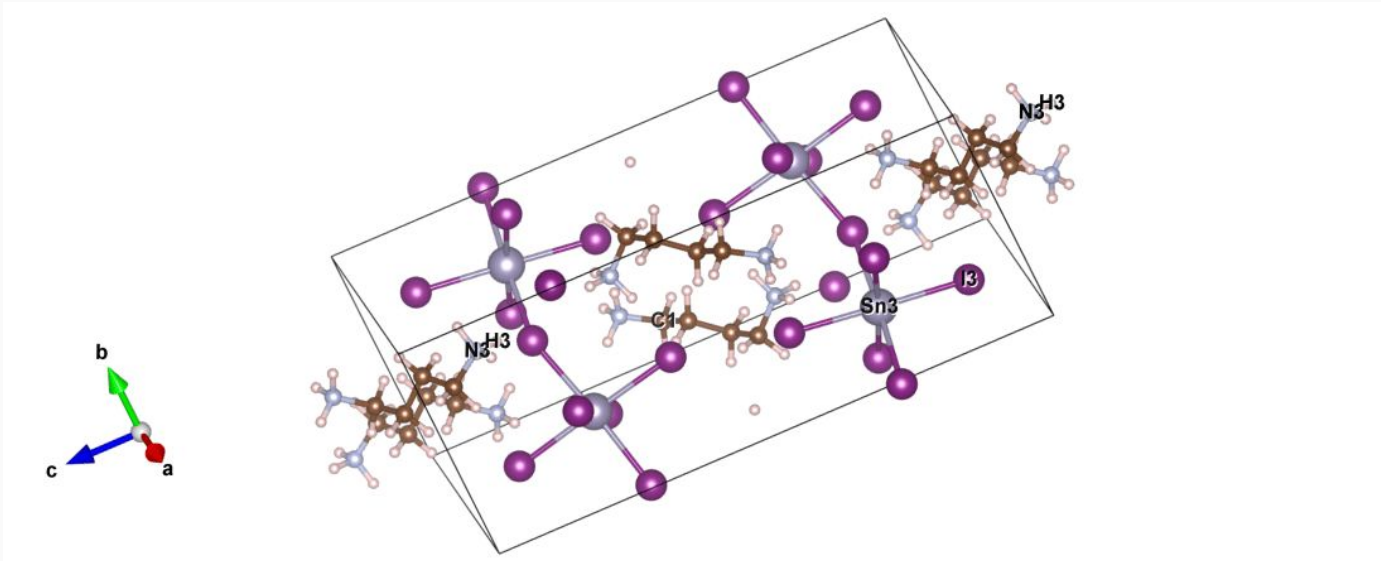


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The dataset: DFT simulations of a hybrid perovskite

The dataset

- System: **hybrid perovskite** (Sn-I cage + putrescine)
- DFT (**VASP**) simulations:
  - NVT, NPT (100 atoms - 300 to 800 K - 1475 cubic angstroms / 0 GPa)
  - 8 runs, 15-20k frames each, 0.5 fs per frame
  - PBE + D3(BJ) functional - Nosé-Hoover / Langevin (γ=10 ps^{-1}) thermostat
- Training data: **position, energy, stress**





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Structural and Dynamical Analysis

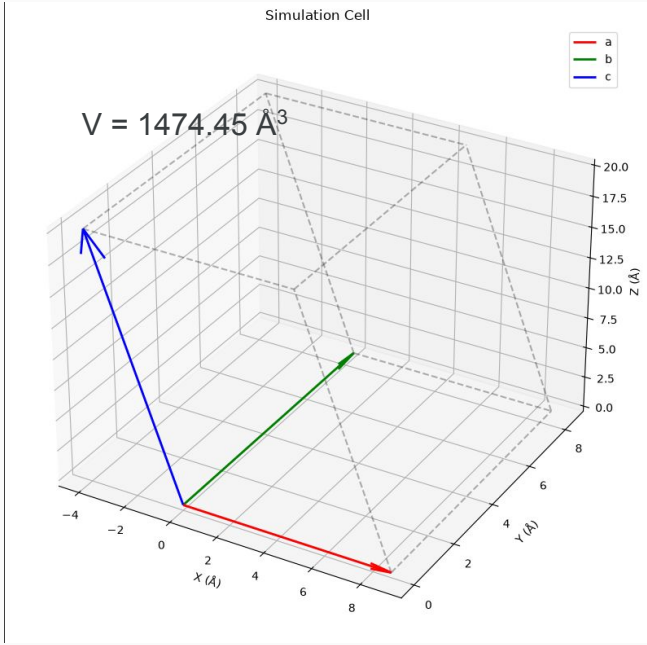
Simulation cell

X = 8.74 Å

Y = 8.42 Å

Z= 20.42 Å

V = 1474.45 Å^3



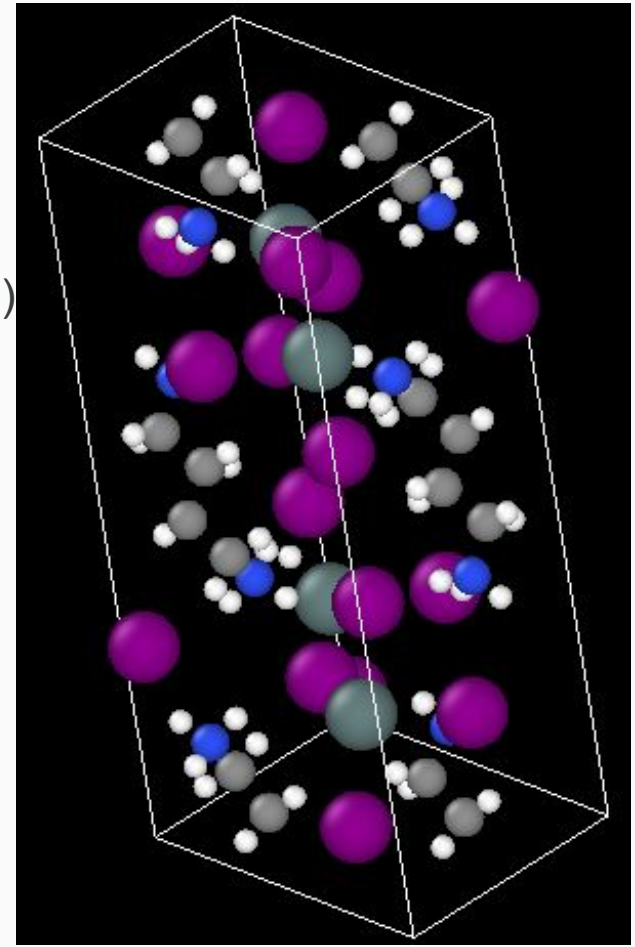
Large grey = Sn (50)

Purple = I (53)

Gray = C (6)

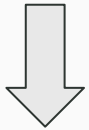
Blue = N (7)

White = H (1)

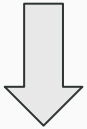


Radial Distribution Function

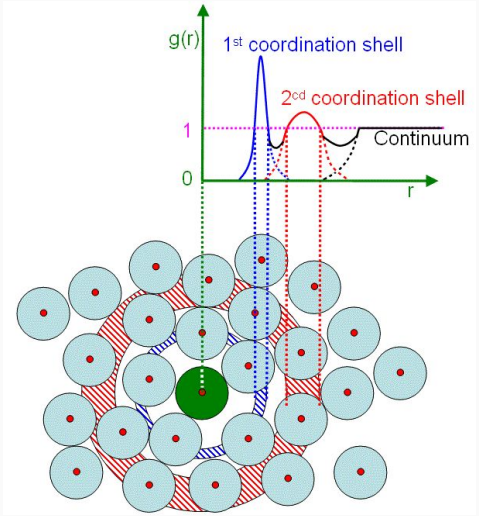
Radial Distribution Function g(r)

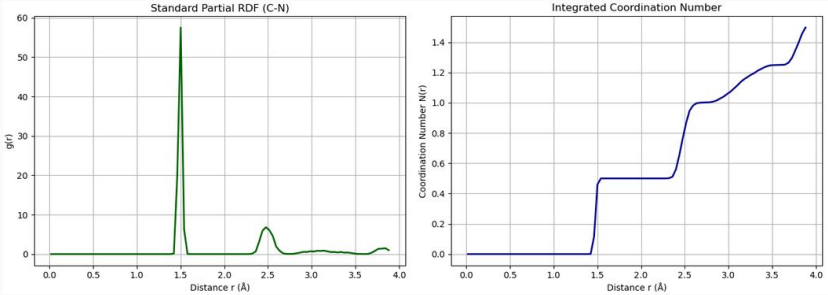


Atomic density



$$N_c(R) = 4\pi\rho \int_0^R r^2 g(r) dr$$

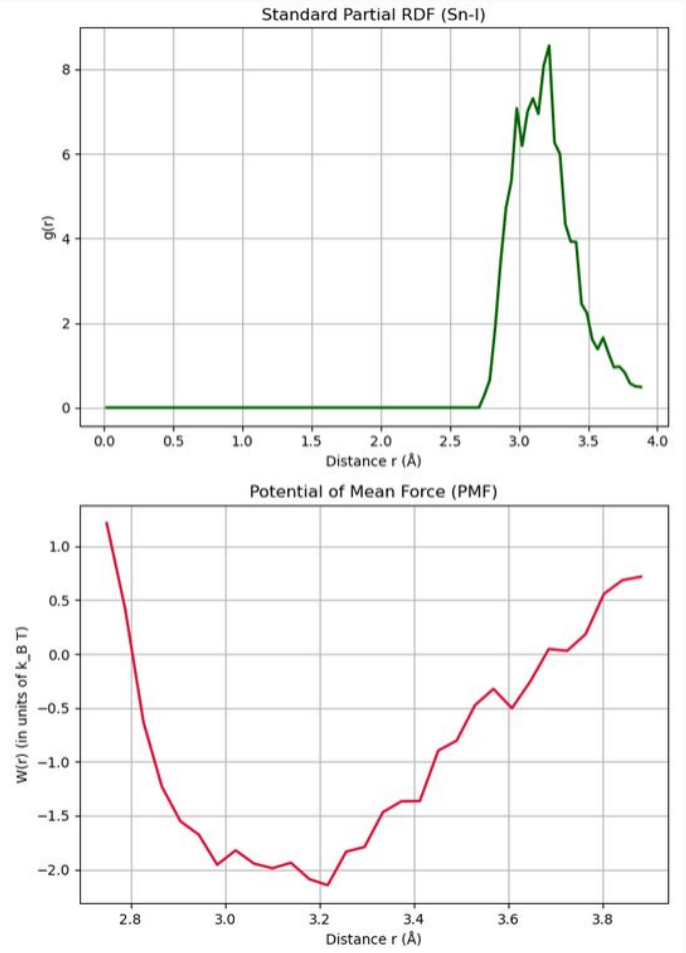




Potential of Mean Force

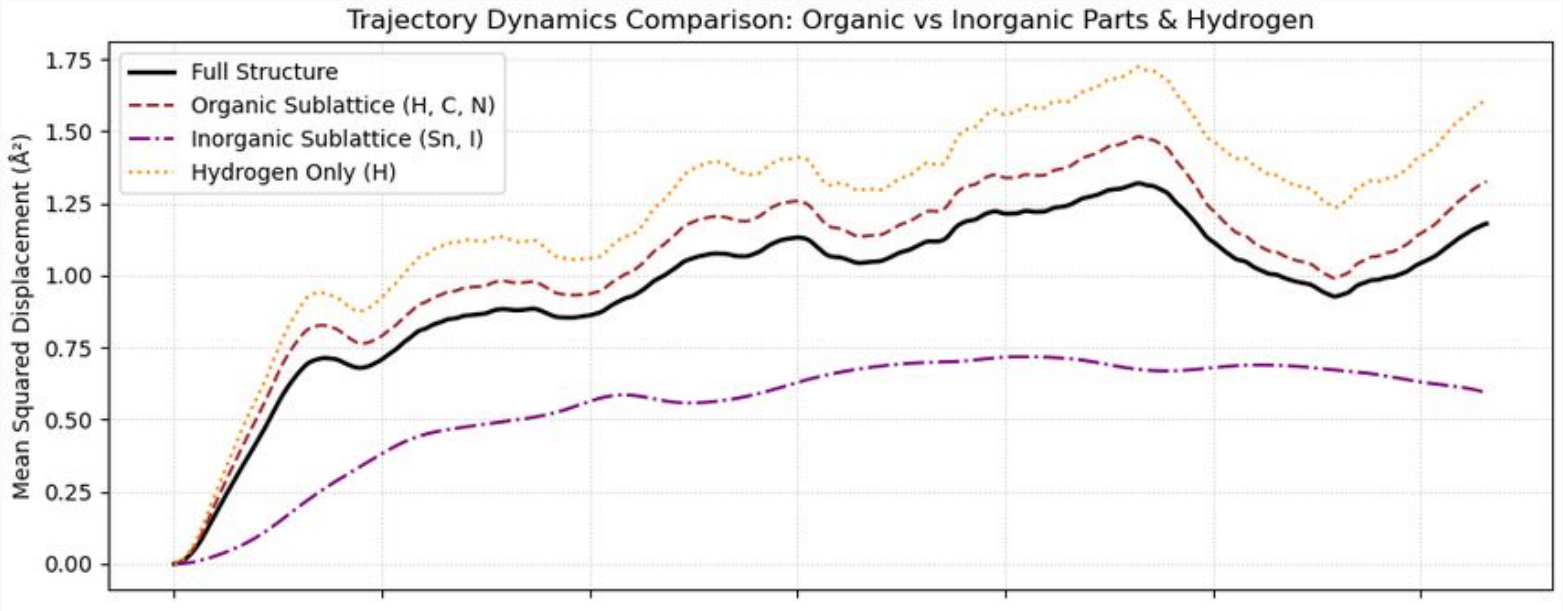
System's free energy:

$$W(r) = -k_B T \ln[g(r)]$$



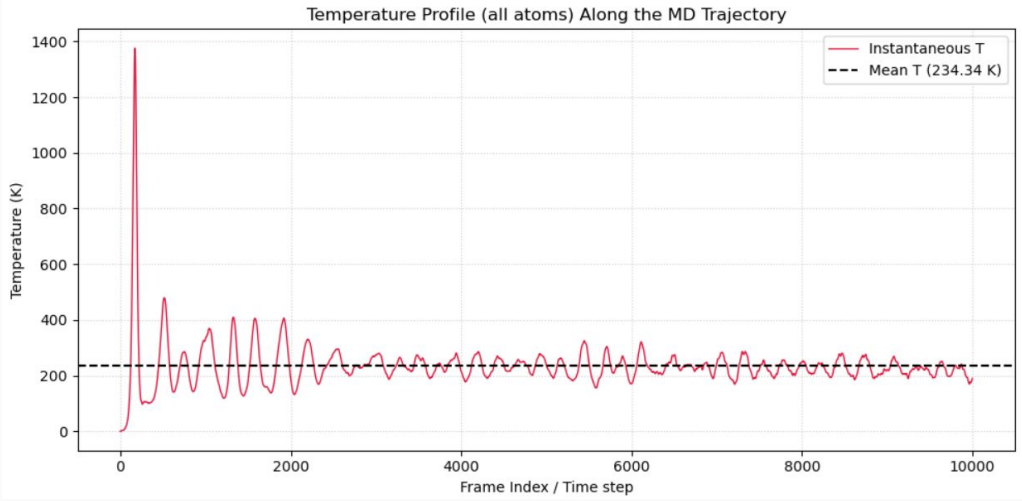
Mean Squared Displacement

$$\mathrm{MSD}(t) = \frac{1}{N} \sum_{i=1}^{N} \left| x_i(t) - x_i(0) \right|^2$$

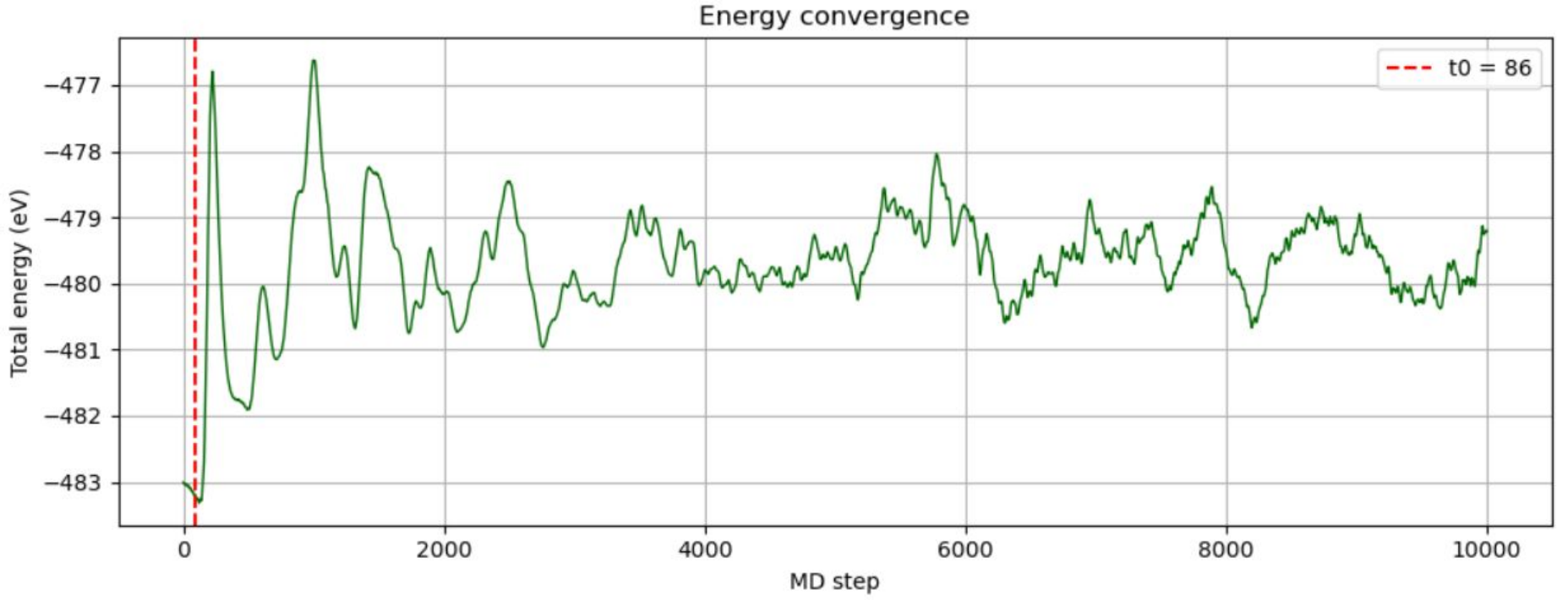


$$E_K(t) = \frac{1}{2} \sum_i m_i \, v_i(t)^2$$

$$T = \frac{2E_k}{N_{\text{dof}} \, k_B}$$



Energy


Energy convergence
t0 = 86
−477
−478
−479
−480
−481
−482
−483
Total energy (eV)
0
2000
4000
6000
8000
10000
MD step

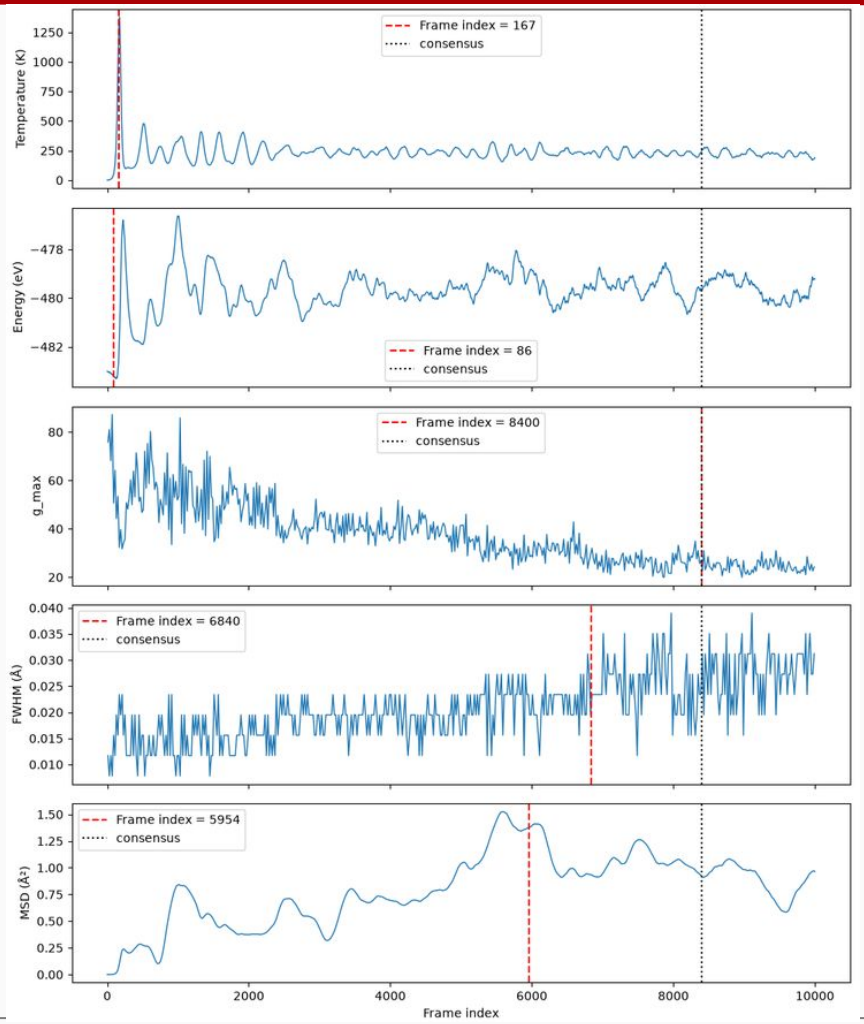
Convergence

Autocorrelation :

$$\rho_t = \frac{\langle (A_n - \mu)(A_{n+t} - \mu) \rangle}{\sigma^2}$$

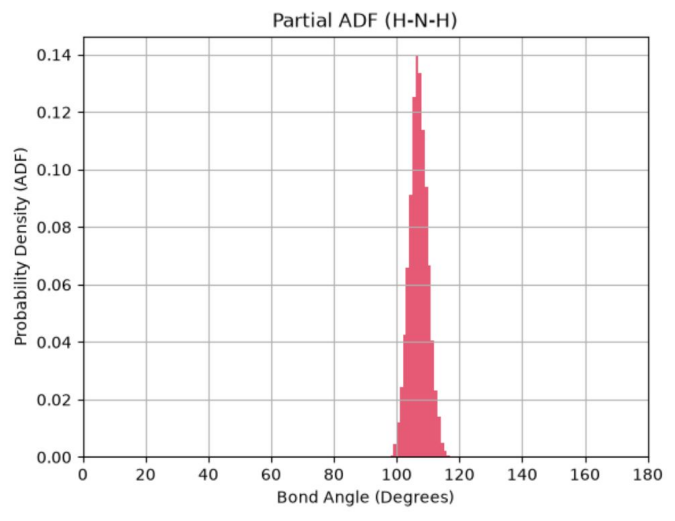
$$g = 1 + 2 \sum_{t=1}^{N-1} \left(1 - \frac{t}{N}\right) \rho_t$$

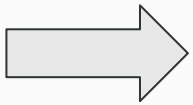
$$N_{\text{eff}} = \frac{N}{g}$$

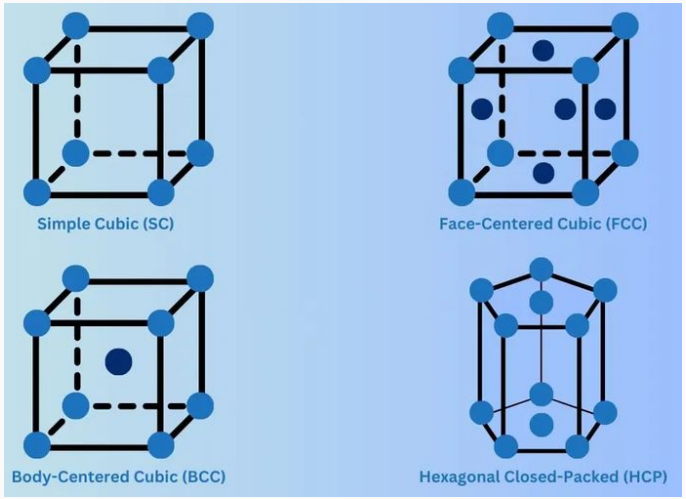


Angular Density Function

The ADF measures the probability distribution of angles formed by triplets of atoms.

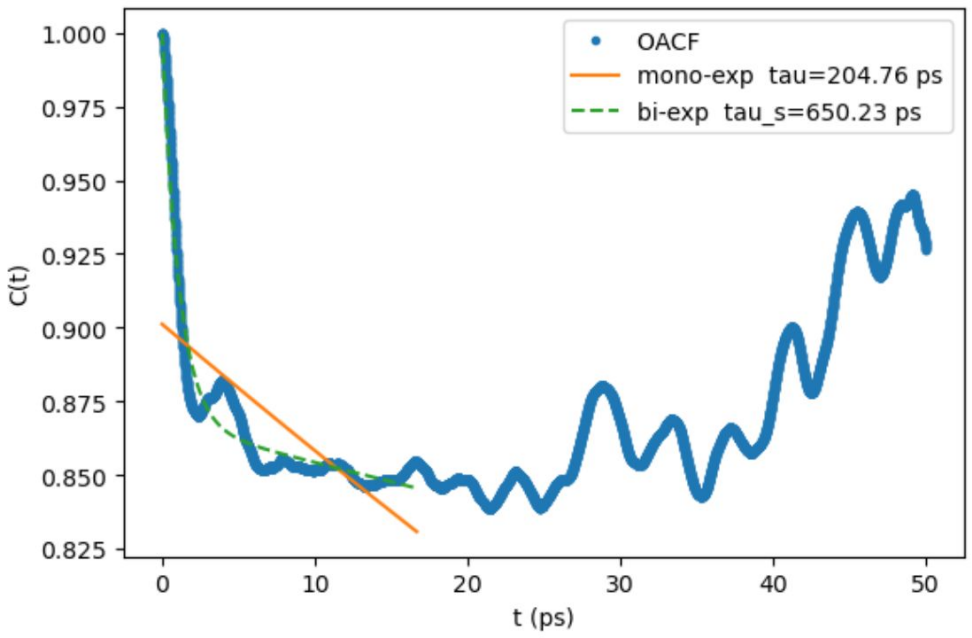






Orientational Autocorrelation Function

$$C(t) = \langle \hat{u}(t_0) \cdot \hat{u}(t_0 + t) \rangle$$

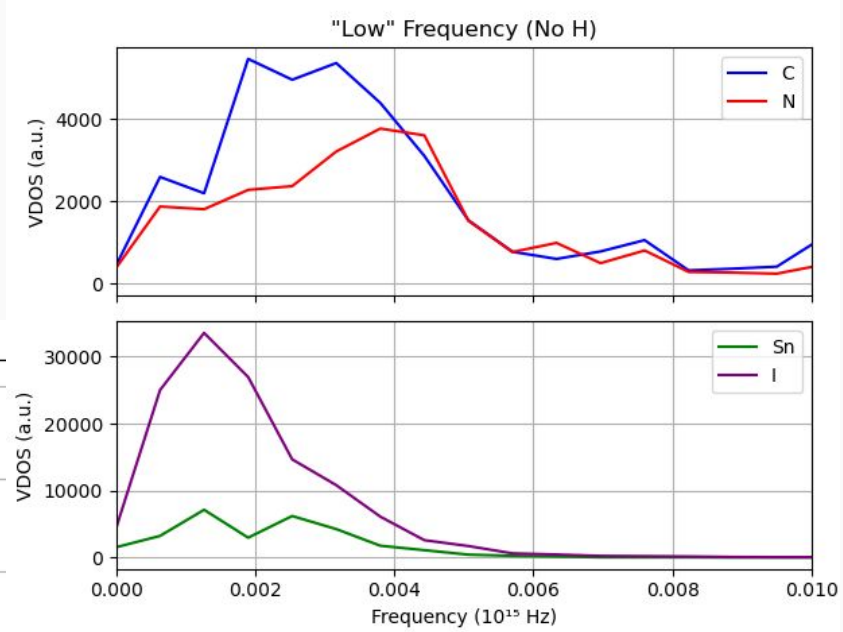


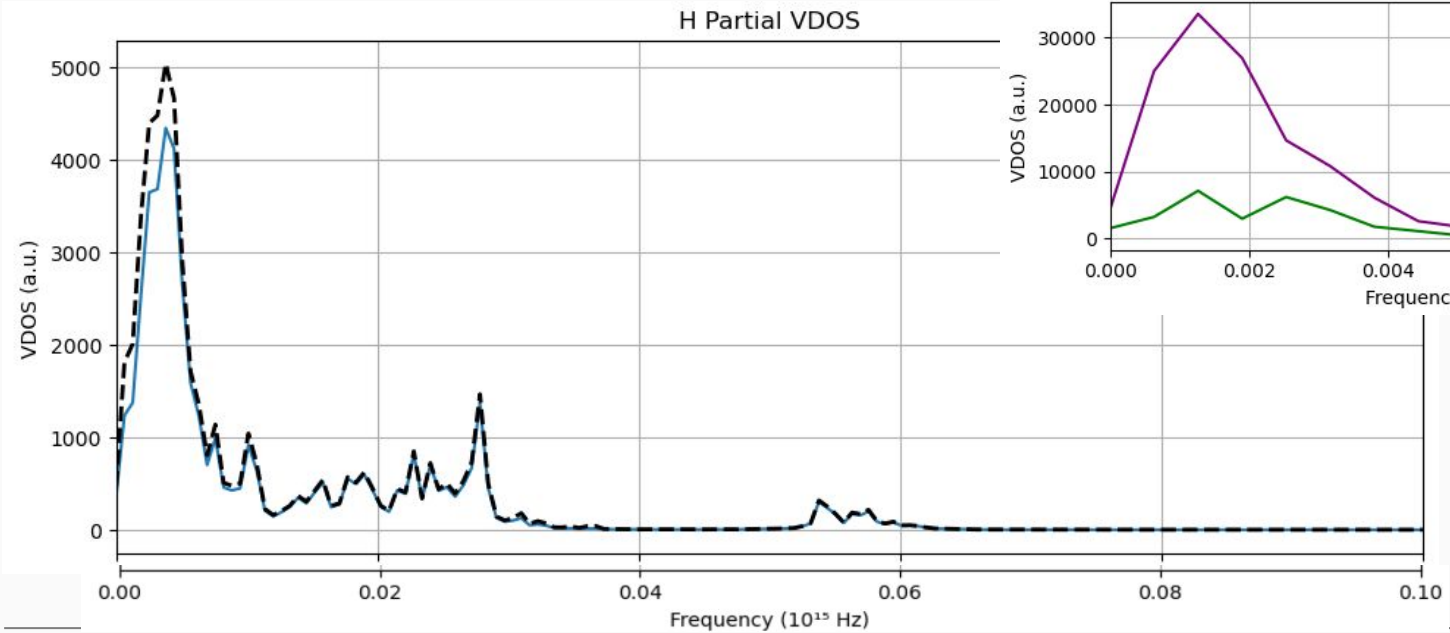
$$C(t) = A\, e^{-t/\tau}$$

$$C(t) = Ae^{-t/\tau_{fast}} + (1 - A)e^{-t/\tau_{slow}}$$

Vibrational Density of States

$$g(\omega) = \int_0^{\infty} \langle \vec{v}(t_0) \cdot \vec{v}(t_0 + t) \rangle \cos(\omega t)\, dt$$







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MACE analysis

MACE: analysis of the library - BASE STRUCTURE

- **Base NN structure: MPNN and GNN**
- **Internal state of the node**
- **Creation of the graph**
- **Message construction and pooling**
- **Update of the internal state**
- **Obtaining physical properties from the internal states**

$$\sigma_i^{(t)} = (\boldsymbol{r}_i, z_i, \boldsymbol{h}_i^{(t)})$$

$$\boldsymbol{m}_i^{(t)} = \bigoplus_{j \in \mathcal{N}(i)} M_t(\sigma_i^{(t)}, \sigma_j^{(t)})$$

$$\boldsymbol{h}_i^{(t+1)} = U_t(\sigma_i^{(t)}, \boldsymbol{m}_i^{(t)})$$

$$E_i = \sum_{t=1}^{T} \mathcal{R}_t(\sigma_i^{(t)})$$

- New message construction mechanism
- Including self-interaction
- E(3) equivariant structure
- Efficient construction of higher order features with smart use of tensor products
- Update of h based on message and state

$$\boldsymbol{m}_i^{(t)} = \sum_j \boldsymbol{u}_1\left(\sigma_i^{(t)}; \sigma_j^{(t)}\right) + \sum_{j_1,j_2} \boldsymbol{u}_2\left(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \sigma_{j_2}^{(t)}\right) + \cdots + \sum_{j_1,\ldots,j_\nu} \boldsymbol{u}_\nu\left(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \ldots, \sigma_{j_\nu}^{(t)}\right)$$

$$A_{i,kl_3m_3}^{(t)} = \sum_{l_1m_1,l_2m_2} C_{l_1m_1,l_2m_2}^{l_3m_3} \sum_{j\in\mathcal{N}(i)} R_{kl_1l_2l_3}^{(t)}(r_{ji}) Y_{l_1}^{m_1}(\hat{\boldsymbol{r}}_{ji}) \sum_{\tilde{k}} W_{k\tilde{k}l_2}^{(t)} h_{j,\tilde{k}l_2m_2}^{(t)}$$

$$\boldsymbol{B}_{i,\eta_\nu kLM}^{(t)} = \sum_{\boldsymbol{lm}} \mathcal{C}_{\eta_\nu,\boldsymbol{lm}}^{LM} \prod_{\xi=1}^{\nu} \sum_{\tilde{k}} w_{k\tilde{k}l_\xi}^{(t)} A_{i,\tilde{k}l_\xi m_\xi}^{(t)}$$

$$m_{i,kLM}^{(t)} = \sum_{\nu} \sum_{\eta_\nu} W_{z_ikL,\eta_\nu}^{(t)} \boldsymbol{B}_{i,\eta_\nu kLM}^{(t)}$$

MACE: analysis of the library - Embedding

- Building of the graph and periodic boundary conditions
- Radial envelope and problem with distance cutoff
- Bessel radial basis and importance
- Spherical harmonics basis
- Chemical embedding

MACE: analysis of the library - Pooling of information

- Encoding of distance information of neighbors
- Encoding of angular informations
- Encoding of chemical information of neighbors
- Producing “halo” of physical information of neighborhood around each atom
- Construction of many-body informations
- Distinction between scalar and directional features
- Conversion of information into physical quantities

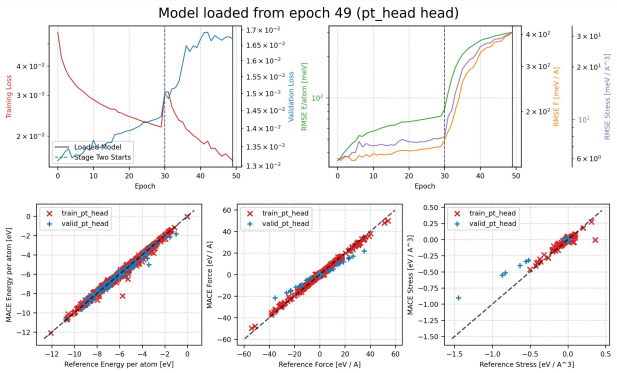
MACE: analysis of the library - Toy model

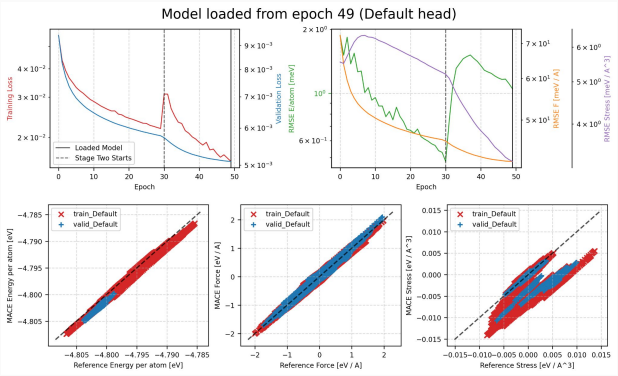
- **Atomic graph builder: PBC Graphing, computing distances, masking out of bounds interaction and self interaction**
- **Radial envelope: smooth radial envelope, ensuring continuity**
- **Bessel radial basis: High dimensional expansion, learnable radial features, no bias, enforce bc**
- **Spherical harmonics: encodes angular info into spherical harmonics, use of e3nn library**
- **Chemical basis: nn.Embedding assigns to each element a learnable list of features**
- **One Particle Basis: pooling of information**
- **Symmetric contraction: responsable to compute many body predictions**
- **Equivariant Update: separates the scalar and directional features in the learning**
- **Atomic readout: update of the states**

MACE OPTIMIZATION: BIAS OF REPULSIVE POTENTIAL

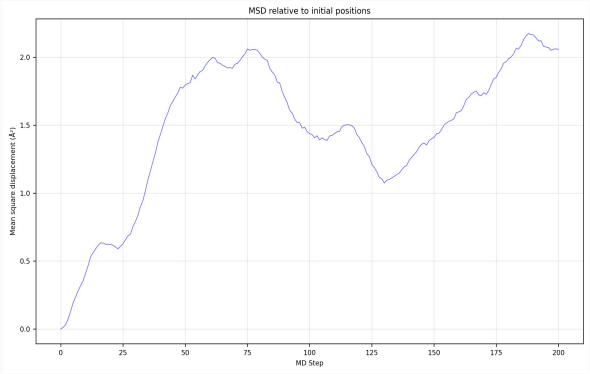
- Ziegler-Biersack-Littmark (ZBL) potential
- Initial problem in the learning and little data about limit cases
- Distance transform
- Rigity of the prior
- Fighting the prior could sometimes lead to worse results

Overfitting issue in base network

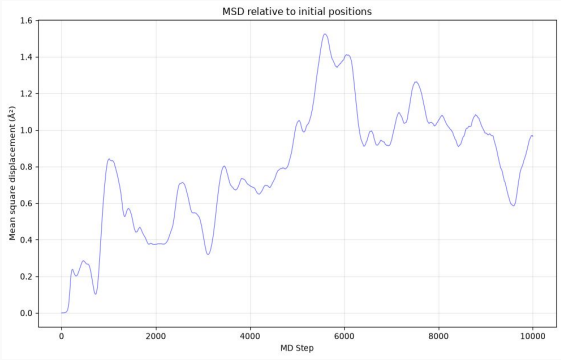




Less stable position



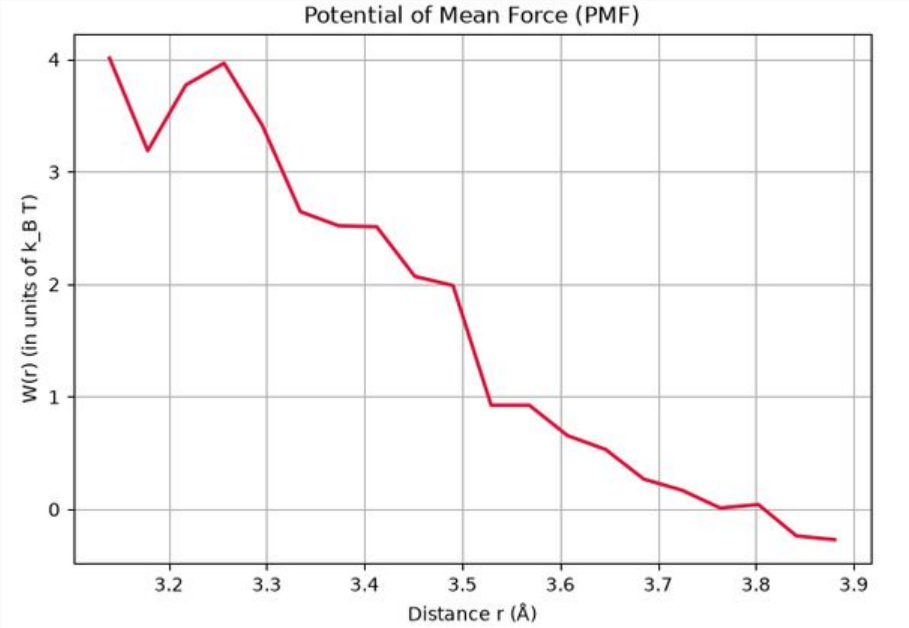
Bias of potential



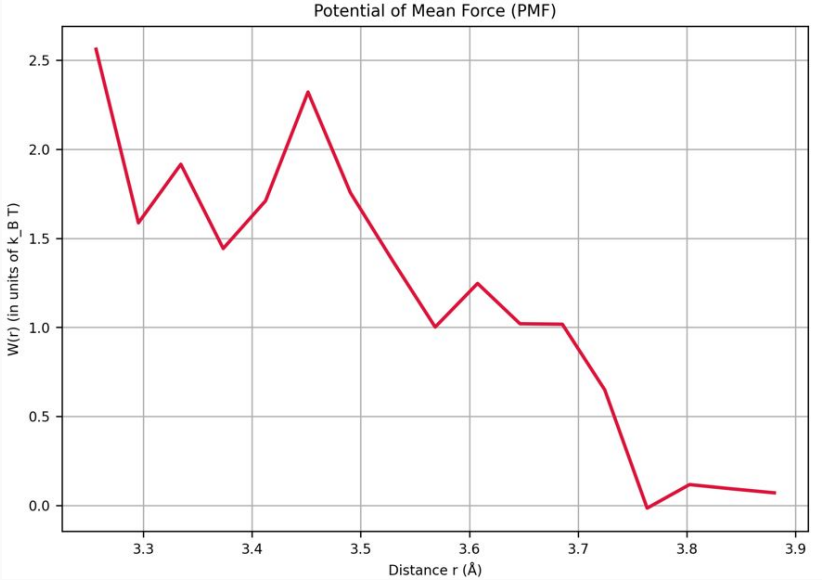
No bias

MACE OPTIMIZATION: BIAS OF REPULSIVE POTENTIAL

Inaccuracies in forces at low distance (PFM I-I)



No bias

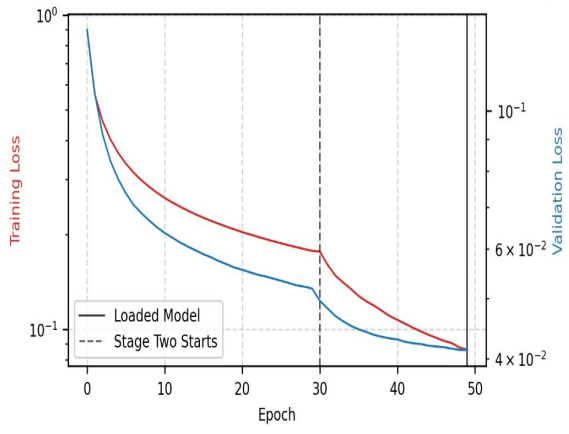


With bias

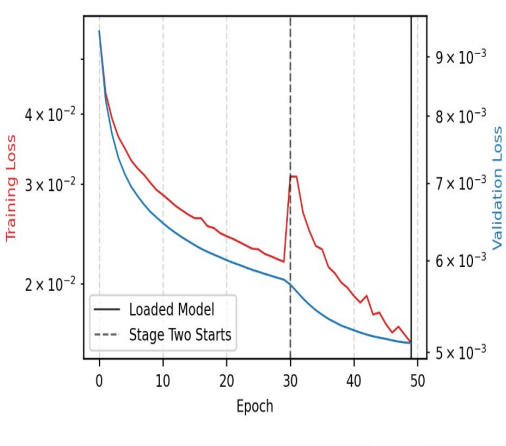
MACE OPTIMIZATION - Effect of Multihead Replay Finetuning

- Pretrained foundation model
- Separation of tasks
- Concurrent batching
- Combination of gradients
- Require more care in the choosing of standard hyperparameters
- Catastrophic forgetting

Slower/Worse convergence to usual error standards



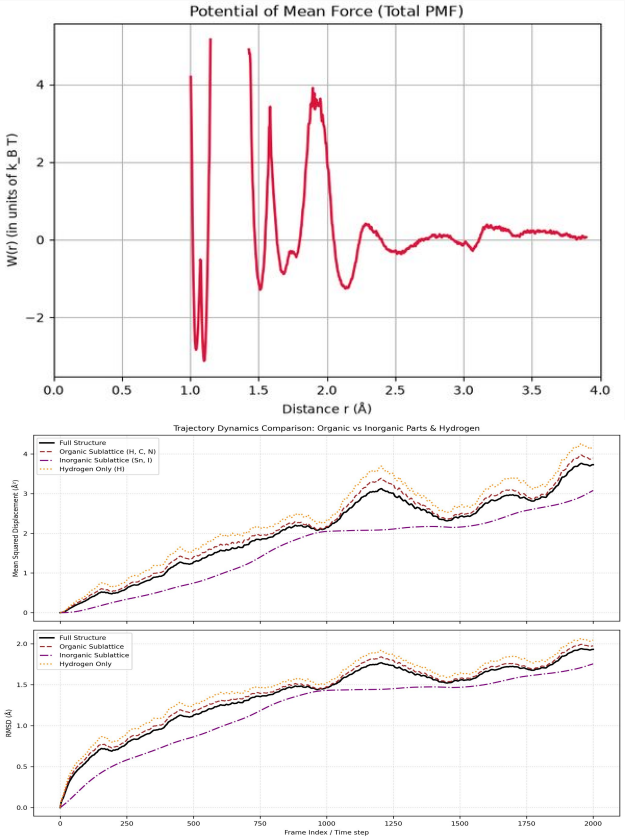
No Multihead Replay Finetuning



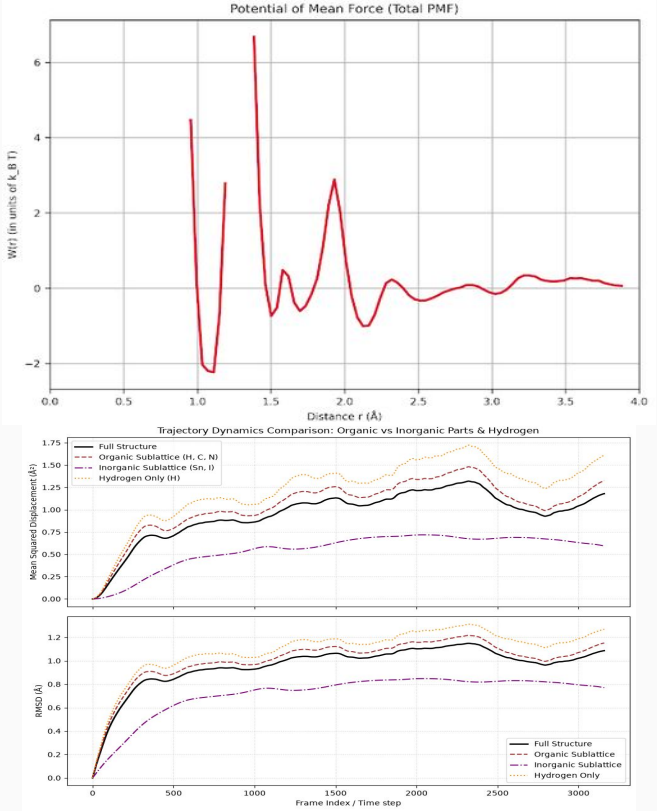
Multihead Replay Finetuning

MACE OPTIMIZATION - Effect of Multihead Replay Finetuning

Less stable dynamics results



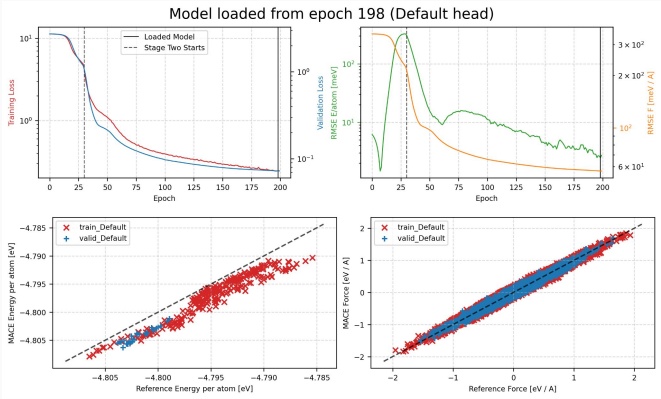
No Multihead Replay Finetuning



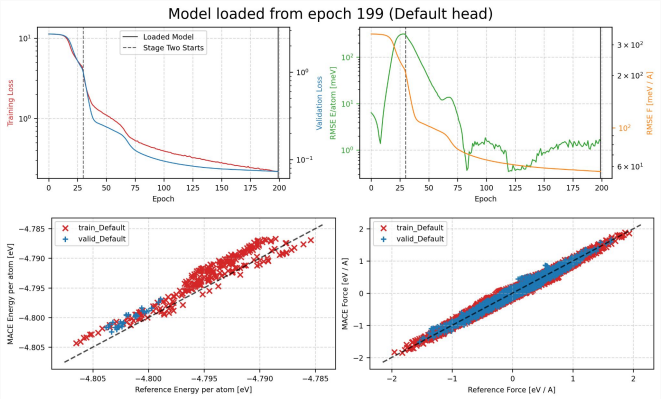
DFT

MACE OPTIMIZATION - Effect of number of n-body interactions

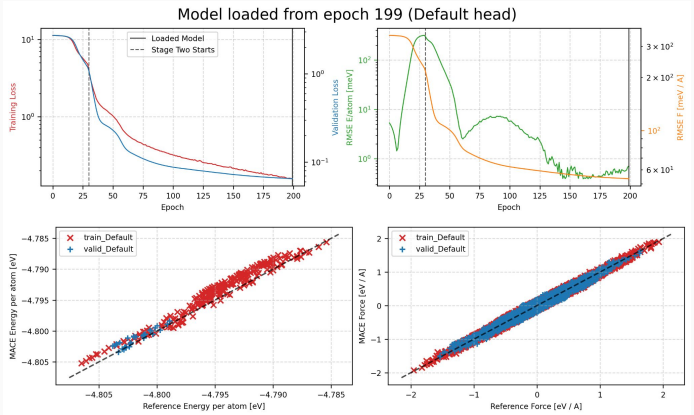
- Higher physical expressivity
- Severe computational bottlenecks and memory spikes
- Exploding feature variance without normalization layer



–correlation=2

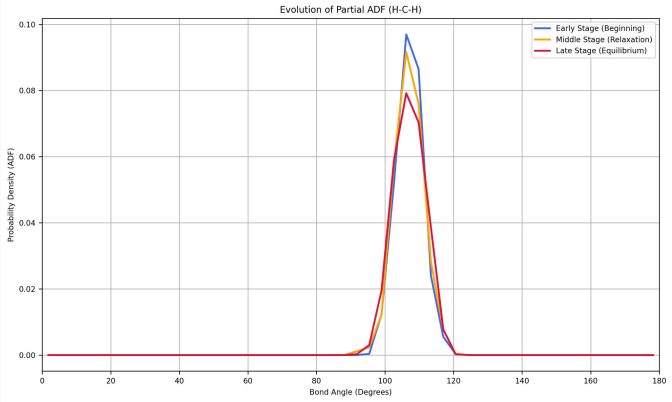


–correlation=3

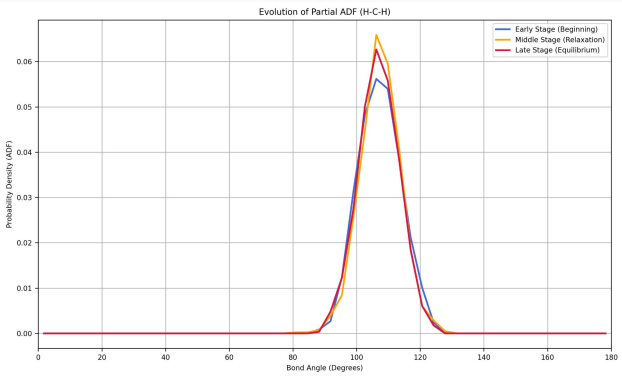


–correlation=4

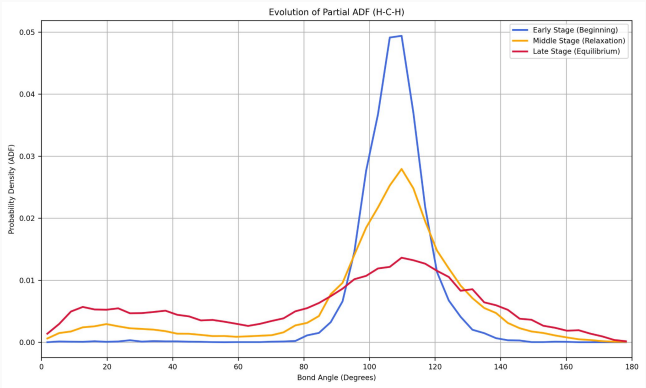
MACE OPTIMIZATION - Effect of number of n-body interactions



–correlation=2



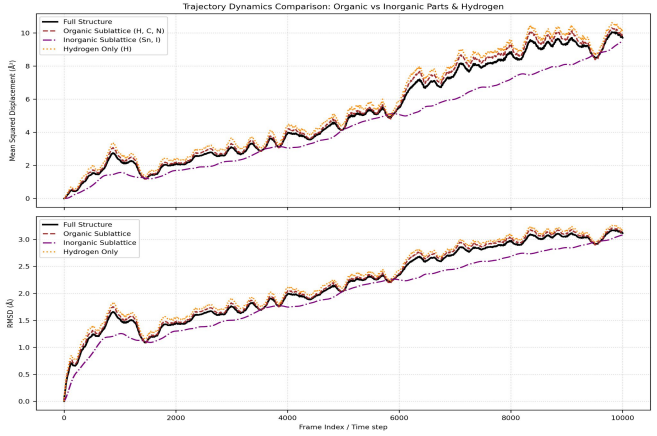
–correlation=3



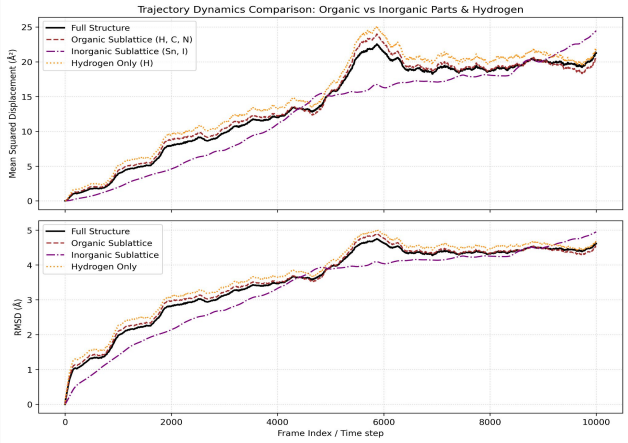
–correlation=4

MACE OPTIMIZATION - Effect of number of n-body interactions

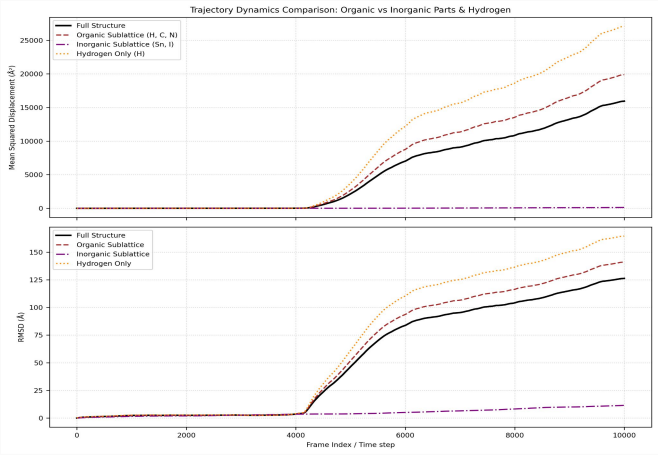
ISSUES WITH BASE MODEL ANALYSIS!
-> probably due to spherical information and no repulsive anchor



–correlation=2



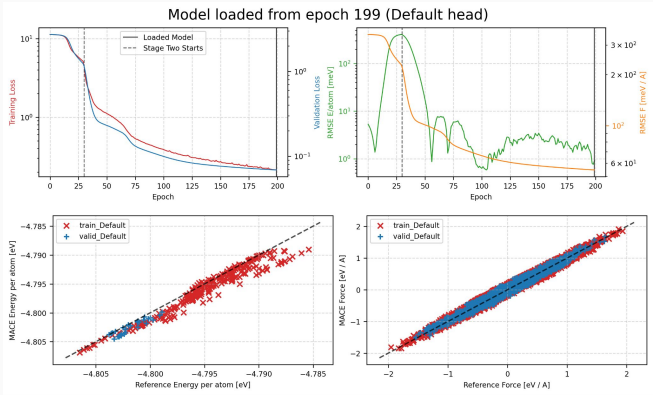
–correlation=3



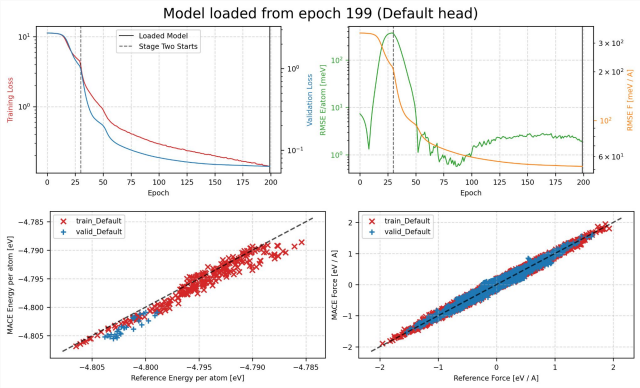
–correlation=4

MACE OPTIMIZATION - Effect of number of n-body interactions, considering angular information (lmax=1)

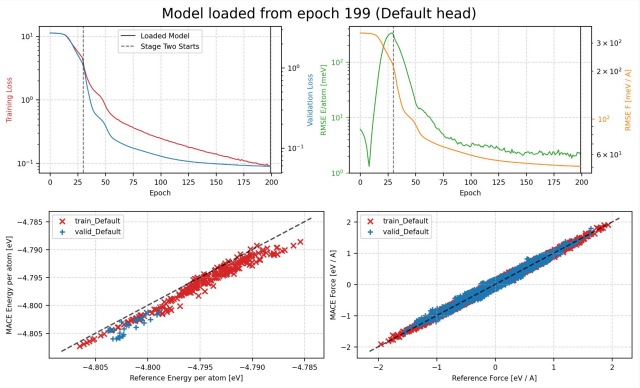
- **Spike in computational effort and training time with –correlation=4 (from approx ~30min to almost 3h)**



–correlation=2



–correlation=3



–correlation=4

MACE OPTIMIZATION - Effect of number of n-body interactions, considering angular information (lmax=1)

- **No significant difference in the analysis on the training convergence, the physical quantities, how much the predictions are distant from the DFT simulation**

[Hydrogen (H)]
Atom # 0: Max distortion 3.06 sigma | 80.7% inside 2-sigma limit
Atom #28: Max distortion 1.69 sigma | 100.0% inside 2-sigma limit
Atom #55: Max distortion 3.62 sigma | 81.3% inside 2-sigma limit
[Carbon (C)]
Atom # 0: Max distortion 4.00 sigma | 93.4% inside 2-sigma limit
Atom # 8: Max distortion 3.54 sigma | 88.1% inside 2-sigma limit
Atom #15: Max distortion 2.53 sigma | 95.9% inside 2-sigma limit
[Nitrogen (N)]
Atom # 0: Max distortion 3.61 sigma | 84.6% inside 2-sigma limit
Atom # 4: Max distortion 4.77 sigma | 80.8% inside 2-sigma limit
Atom # 7: Max distortion 4.16 sigma | 90.3% inside 2-sigma limit
[Tin (Sn)]
Atom # 0: Max distortion 5.08 sigma | 91.7% inside 2-sigma limit
Atom # 2: Max distortion 5.18 sigma | 91.6% inside 2-sigma limit
Atom # 3: Max distortion 4.69 sigma | 88.3% inside 2-sigma limit
[Iodine (I)]
Atom # 0: Max distortion 4.68 sigma | 90.9% inside 2-sigma limit
Atom # 8: Max distortion 2.88 sigma | 90.8% inside 2-sigma limit
Atom #15: Max distortion 3.68 sigma | 78.9% inside 2-sigma limit

–correlation=2

[Hydrogen (H)]
Atom # 0: Max distortion 3.27 sigma | 84.1% inside 2-sigma limit
Atom #28: Max distortion 1.51 sigma | 100.0% inside 2-sigma limit
Atom #55: Max distortion 3.40 sigma | 74.9% inside 2-sigma limit
[Carbon (C)]
Atom # 0: Max distortion 4.04 sigma | 93.1% inside 2-sigma limit
Atom # 8: Max distortion 2.75 sigma | 91.3% inside 2-sigma limit
Atom #15: Max distortion 2.61 sigma | 95.7% inside 2-sigma limit
[Nitrogen (N)]
Atom # 0: Max distortion 3.85 sigma | 85.6% inside 2-sigma limit
Atom # 4: Max distortion 4.22 sigma | 81.3% inside 2-sigma limit
Atom # 7: Max distortion 3.90 sigma | 87.7% inside 2-sigma limit
[Tin (Sn)]
Atom # 0: Max distortion 4.36 sigma | 84.8% inside 2-sigma limit
Atom # 2: Max distortion 4.45 sigma | 89.7% inside 2-sigma limit
Atom # 3: Max distortion 4.21 sigma | 82.8% inside 2-sigma limit
[Iodine (I)]
Atom # 0: Max distortion 3.18 sigma | 94.2% inside 2-sigma limit
Atom # 8: Max distortion 3.22 sigma | 89.7% inside 2-sigma limit
Atom #15: Max distortion 3.51 sigma | 86.0% inside 2-sigma limit

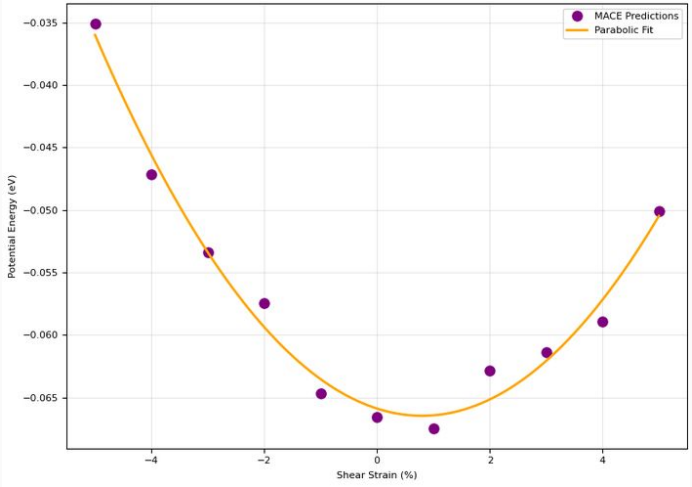
–correlation=3

[Hydrogen (H)]
Atom # 0: Max distortion 2.98 sigma | 93.8% inside 2-sigma limit
Atom #28: Max distortion 1.95 sigma | 100.0% inside 2-sigma limit
Atom #55: Max distortion 3.09 sigma | 88.5% inside 2-sigma limit
[Carbon (C)]
Atom # 0: Max distortion 2.55 sigma | 97.1% inside 2-sigma limit
Atom # 8: Max distortion 3.78 sigma | 93.2% inside 2-sigma limit
Atom #15: Max distortion 2.81 sigma | 96.5% inside 2-sigma limit
[Nitrogen (N)]
Atom # 0: Max distortion 3.66 sigma | 81.3% inside 2-sigma limit
Atom # 4: Max distortion 4.15 sigma | 73.9% inside 2-sigma limit
Atom # 7: Max distortion 3.54 sigma | 91.8% inside 2-sigma limit
[Tin (Sn)]
Atom # 0: Max distortion 2.88 sigma | 94.6% inside 2-sigma limit
Atom # 2: Max distortion 2.52 sigma | 94.8% inside 2-sigma limit
Atom # 3: Max distortion 3.91 sigma | 85.0% inside 2-sigma limit
[Iodine (I)]
Atom # 0: Max distortion 4.22 sigma | 90.8% inside 2-sigma limit
Atom # 8: Max distortion 3.73 sigma | 89.4% inside 2-sigma limit
Atom #15: Max distortion 3.37 sigma | 83.1% inside 2-sigma limit

–correlation=4

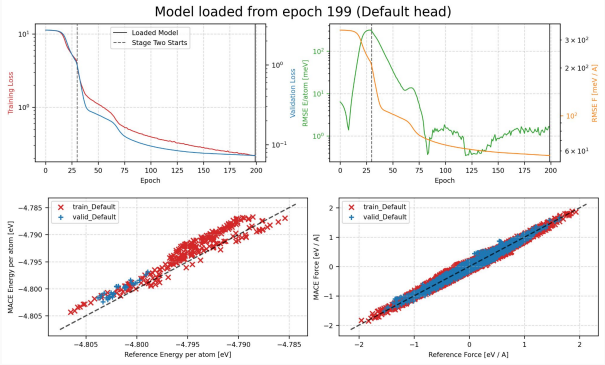
MACE OPTIMIZATION - Effect of number of n-body interactions with lmax=1, shear strain analysis

- Resistance to shape change
- Simulating 0K and finding minimum of potential
- FrechetCellFilter
- L-BFGS algorithm
- moving the atoms and resizing the box until the maximum force pushing on any single atom drops below 0.05 eV/Å
- **cell[0, 1] += strain * cell[1, 1] :** distort the structur, applying the transformation, finding the new configuration for atoms
- Quadratic fit and taking the second order coefficient
- Decreasing stiffness passing from 3-body to 4-body interaction but rising again with 5-body -> supposed to be linked to the fact that Sn and 4 I are considered all together

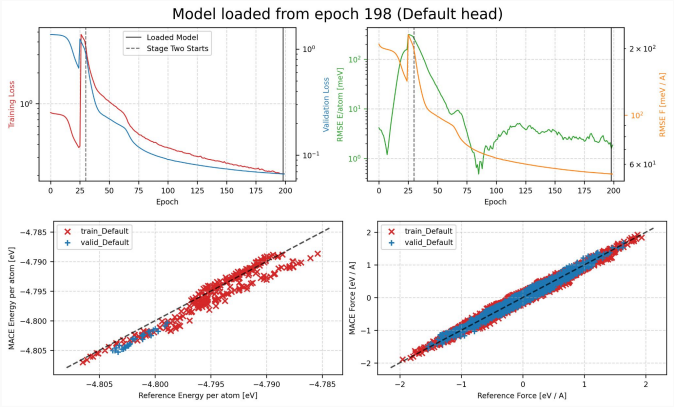


MACE OPTIMIZATION - Effect of changing the number of message passing layers

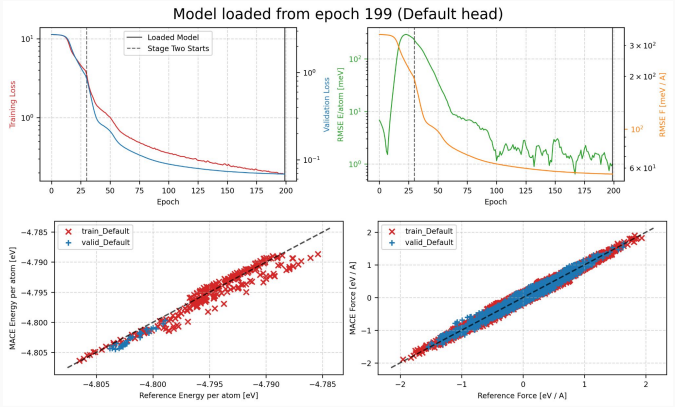
- Higher receptive field as the –num_interaction parameter rises
- Possible overfitting
- Computational complexity
- No evident difference in the MACE training diagnostic and comparison in energies and forces with the dft simulation



–num_inter=2



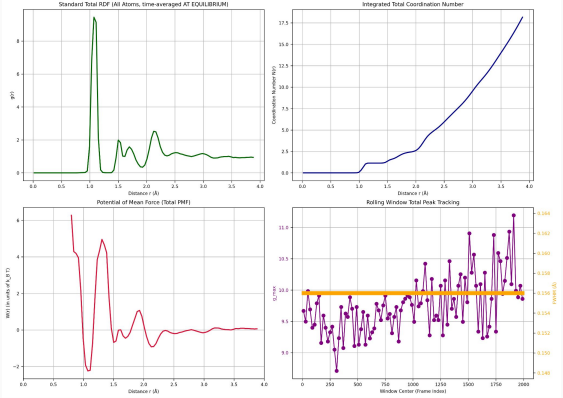
–num_inter=3



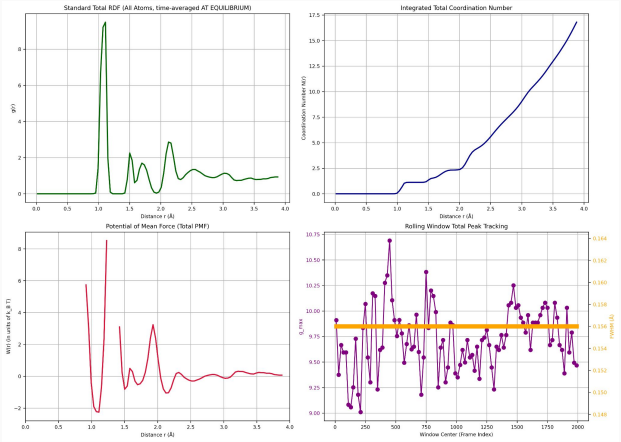
–num_inter=4

MACE OPTIMIZATION - Effect of changing the number of message passing layers

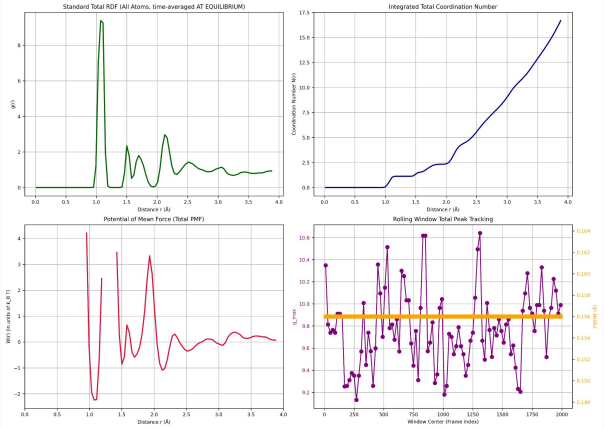
- No visible changes in the analysis
- Slight changes in vibrational spectrum and in force at low distance
- Possibly more impactful with more angular information



–num_inter=2



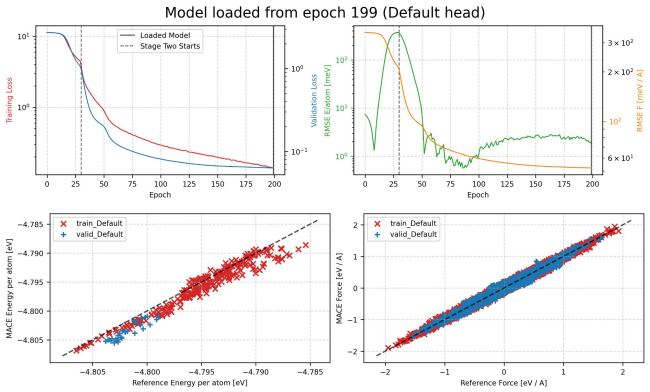
–num_inter=3



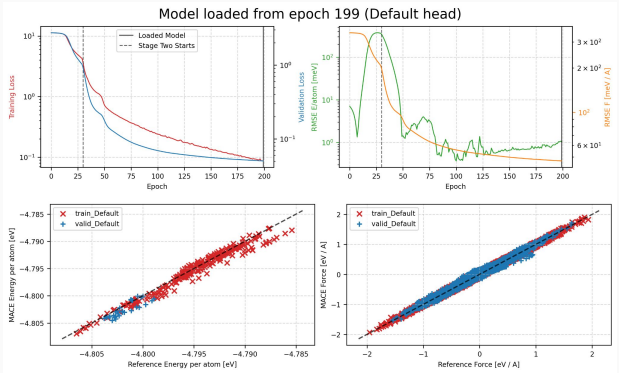
–num_inter=4

MACE OPTIMIZATION - Effect of changing the number of message passing layers - adding spherical information

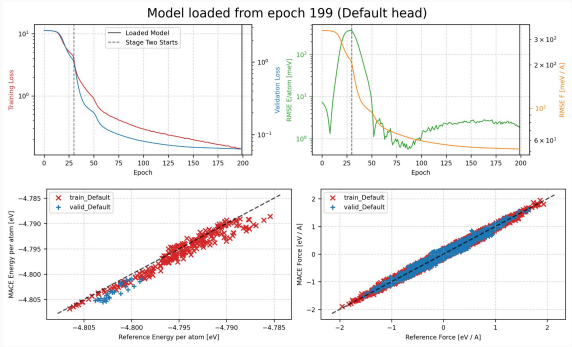
- **Almost identical training diagnostic, energy and force determination**
- **Higher computational effort observed for –num_inter higher then 2**
- **All physical analysis shows no difference**



–num_inter=2



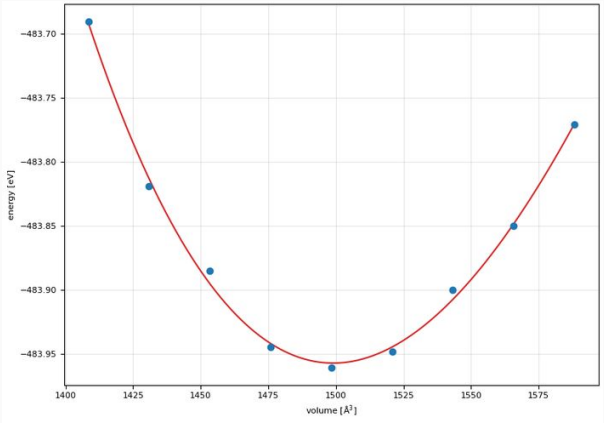
–num_inter=3



–num_inter=4

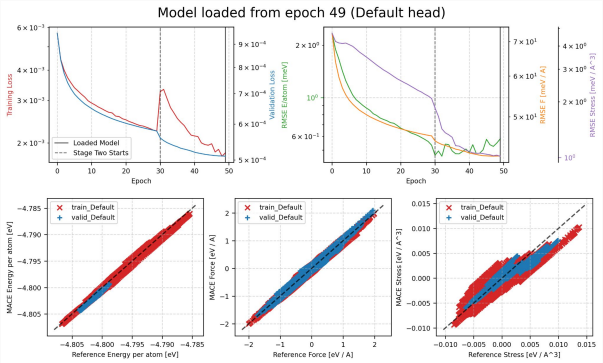
MACE OPTIMIZATION - Effect of changing the number of message passing layers - Equation of state analysis

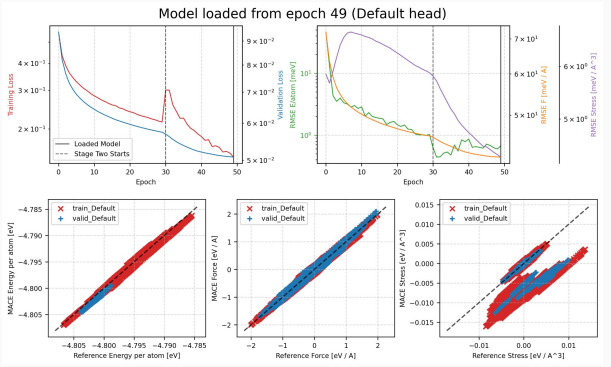
- Simulating 0K and finding minimum of potential
- FrechetCellFilter
- L-BFGS algorithm
- moving the atoms and resizing the box until the maximum force pushing on any single atom drops below 0.05 eV/Å
- crushing and stretching the crystal to measure how strongly it resists changing its volume
- atoms_strained.set_cell with scale_atoms=True, find the new equilibrium (computing energy)
- Fit with Birch-Murnaghan function v0, e0, B = eos.fit()
- Comparing different values of num_interaction -> same results
- Hypothesized that maybe more spherical harmonics could make a difference but it is outside of the computation range

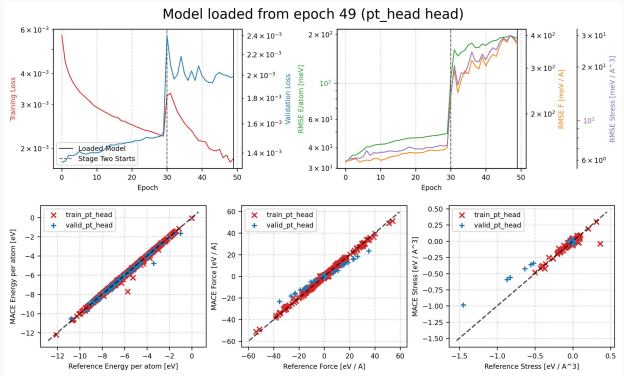


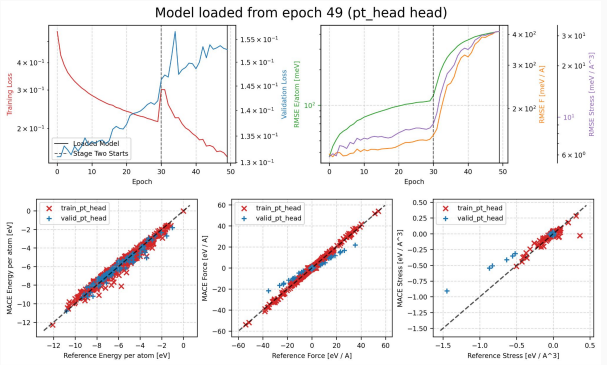
MACE OPTIMIZATION - Effect of changing the relevance of force

- **Imprecisions in force determination at with low force and high weight in pt_head, while energy determination remains accurate, with better results at high force weight**









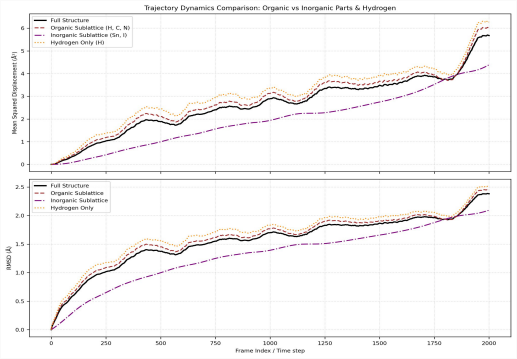
force weight: 10.0

force weight: 1000.0

MACE OPTIMIZATION - Effect of changing the relevance of force

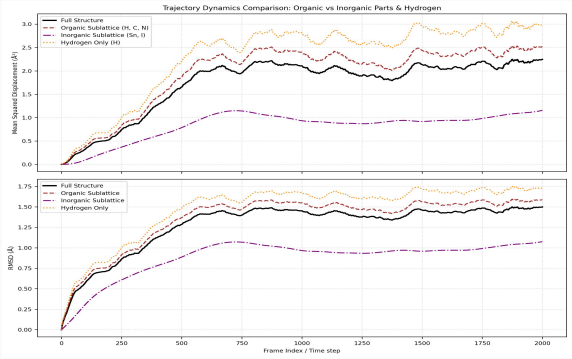
- **MSD more aligned with DFT analysis with high force weight**
- **Overall similar percentage of reasonable predictions but with low force weight higher outsiders**
- **Higher bulk modulus for 1000.0, more aligned with previous analysis**

[Hydrogen (H)]
Atom # 0: Max distortion 3.32 sigma | 82.2% inside 2-sigma limit
Atom #28: Max distortion 2.18 sigma | 99.7% inside 2-sigma limit
Atom #55: Max distortion 3.78 sigma | 75.1% inside 2-sigma limit
[Carbon (C)]
Atom # 0: Max distortion 3.35 sigma | 92.1% inside 2-sigma limit
Atom # 8: Max distortion 3.48 sigma | 92.7% inside 2-sigma limit
Atom #15: Max distortion 3.18 sigma | 95.2% inside 2-sigma limit
[Nitrogen (N)]
Atom # 0: Max distortion 4.23 sigma | 84.2% inside 2-sigma limit
Atom # 4: Max distortion 4.24 sigma | 80.5% inside 2-sigma limit
Atom # 7: Max distortion 3.37 sigma | 89.7% inside 2-sigma limit
[Tin (Sn)]
Atom # 0: Max distortion 2.84 sigma | 90.1% inside 2-sigma limit
Atom # 2: Max distortion 4.07 sigma | 80.6% inside 2-sigma limit
Atom # 3: Max distortion 3.30 sigma | 93.2% inside 2-sigma limit
[Iodine (I)]
Atom # 0: Max distortion 2.43 sigma | 96.2% inside 2-sigma limit
Atom # 8: Max distortion 2.77 sigma | 98.8% inside 2-sigma limit
Atom #15: Max distortion 3.40 sigma | 80.9% inside 2-sigma limit



force weight: 10.0

[Hydrogen (H)]
Atom # 0: Max distortion 3.16 sigma | 82.7% inside 2-sigma limit
Atom #28: Max distortion 2.09 sigma | 99.4% inside 2-sigma limit
Atom #55: Max distortion 3.86 sigma | 77.2% inside 2-sigma limit
[Carbon (C)]
Atom # 0: Max distortion 2.85 sigma | 91.9% inside 2-sigma limit
Atom # 8: Max distortion 2.49 sigma | 93.4% inside 2-sigma limit
Atom #15: Max distortion 3.47 sigma | 90.8% inside 2-sigma limit
[Nitrogen (N)]
Atom # 0: Max distortion 2.59 sigma | 85.8% inside 2-sigma limit
Atom # 4: Max distortion 4.78 sigma | 81.2% inside 2-sigma limit
Atom # 7: Max distortion 4.18 sigma | 90.0% inside 2-sigma limit
[Tin (Sn)]
Atom # 0: Max distortion 2.98 sigma | 88.0% inside 2-sigma limit
Atom # 2: Max distortion 2.74 sigma | 84.2% inside 2-sigma limit
Atom # 3: Max distortion 2.61 sigma | 89.1% inside 2-sigma limit
[Iodine (I)]
Atom # 0: Max distortion 1.93 sigma | 100.0% inside 2-sigma limit
Atom # 8: Max distortion 2.38 sigma | 99.1% inside 2-sigma limit
Atom #15: Max distortion 3.74 sigma | 79.5% inside 2-sigma limit

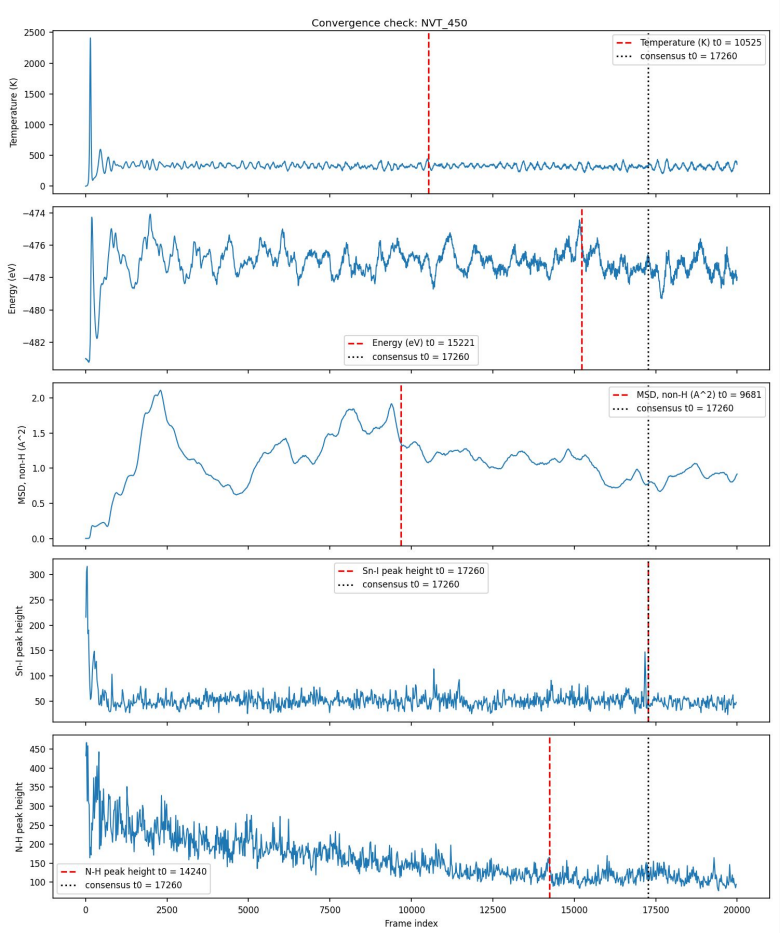


force weight: 1000.0

MACE OPTIMIZATION - Effect of changing the dataset (converged or not)

SETUP

- Assessing VASP convergence: conservative tail-band heuristics

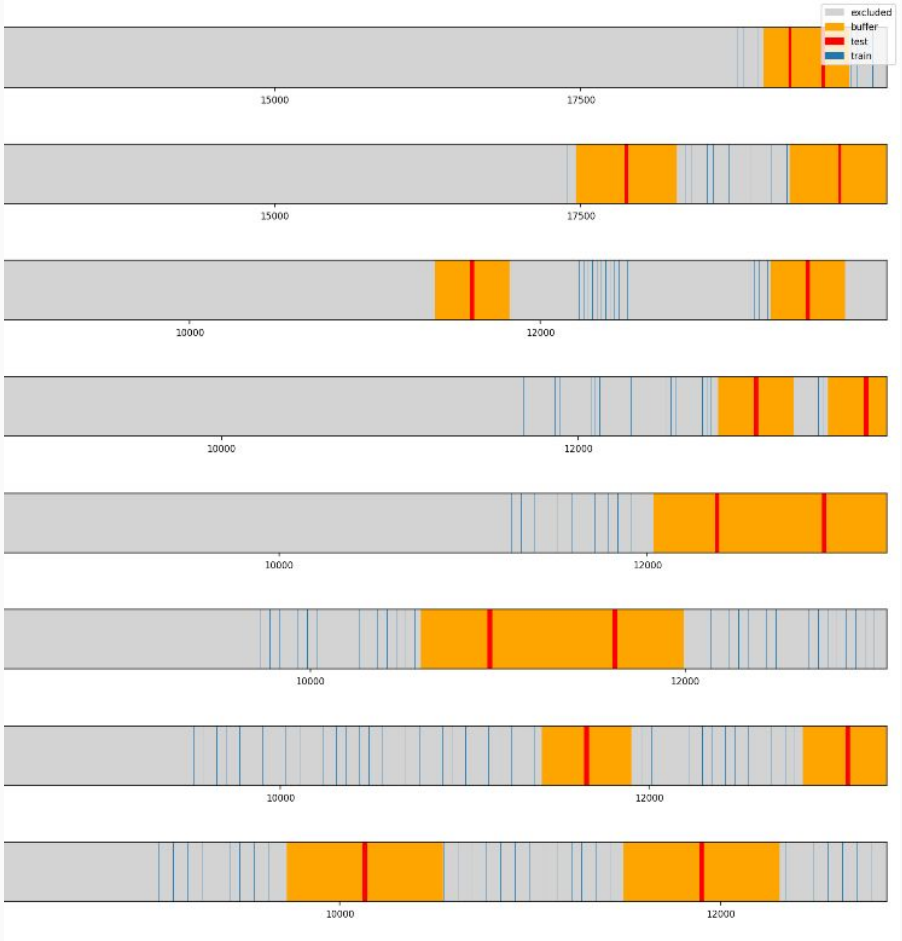


Github: notebook and script

MACE OPTIMIZATION - Effect of changing the dataset (converged or not)

SETUP

- Test set: same, converged frames. Large buffer from training
- Training set: strided frames. Full versus converged-only. Both 518 frames



Github: notebook and script

MACE OPTIMIZATION - Effect of changing the dataset (converged or not)

RESULTS

- Limitations: one run, little data, simple model, slow convergence
- Observed: no significant difference:

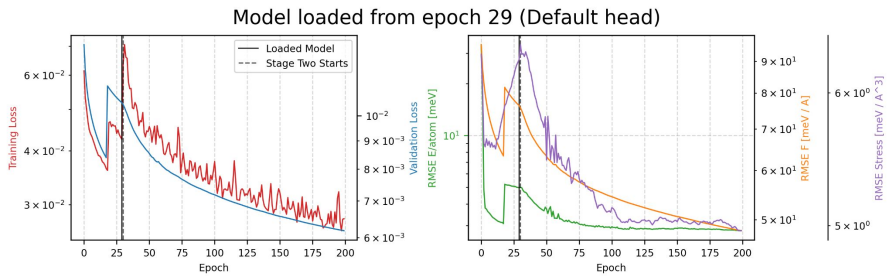
	Energy RMSE (meV)	Energy R-2 score (%)	Forces RMSE (meV/A)	Forces R-2 score (%)
Converged only data	**255.2**	**99.07**	53.8	99.57
Full data	268.9	98.96	**53.6**	99.57

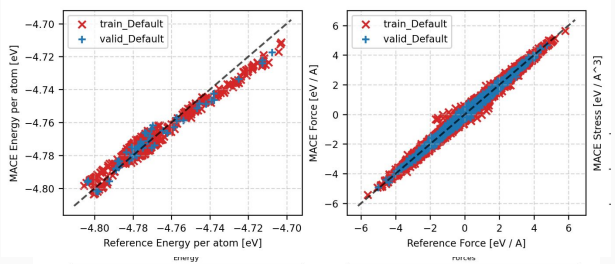
Github: notebook and script

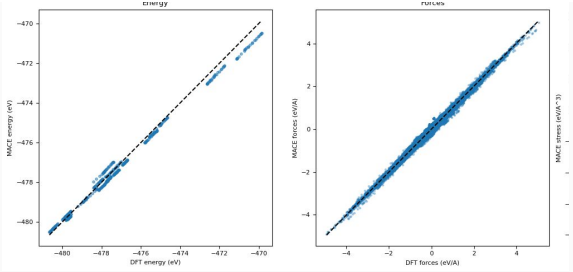
MACE OPTIMIZATION - Effect of changing the dataset (converged or not)

RESULTS

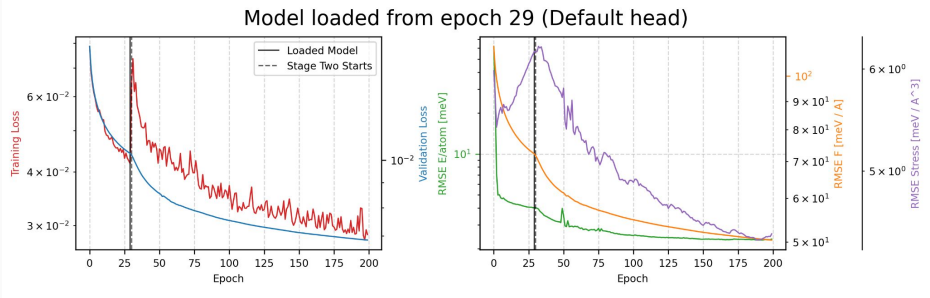
CONVERGED-ONLY DATA

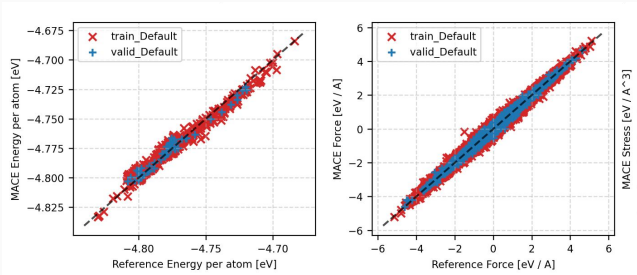


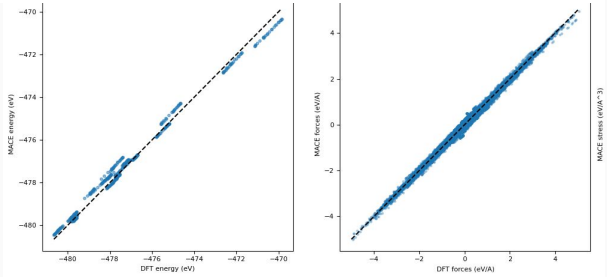




FULL DATA

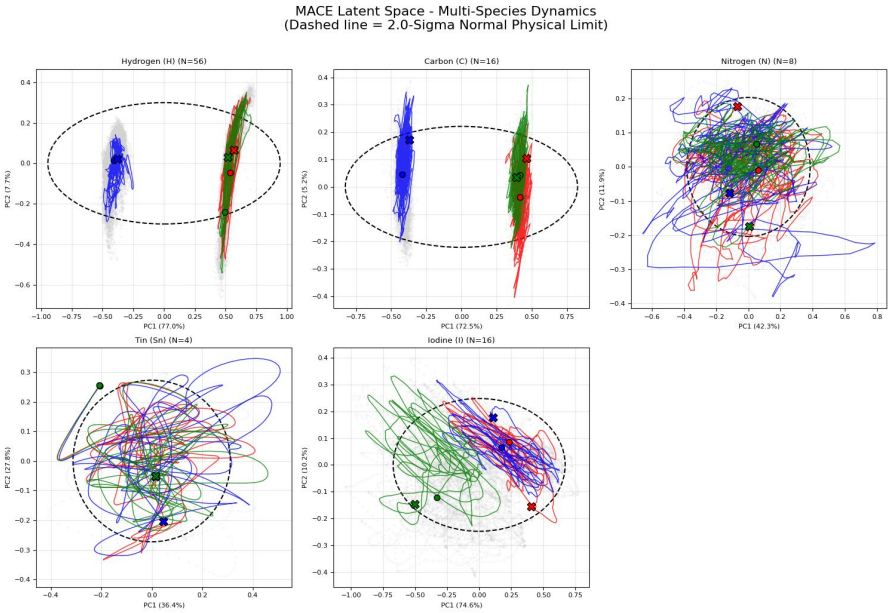






Github: notebook and script

MACE - Quality of the fit: latent space analysis



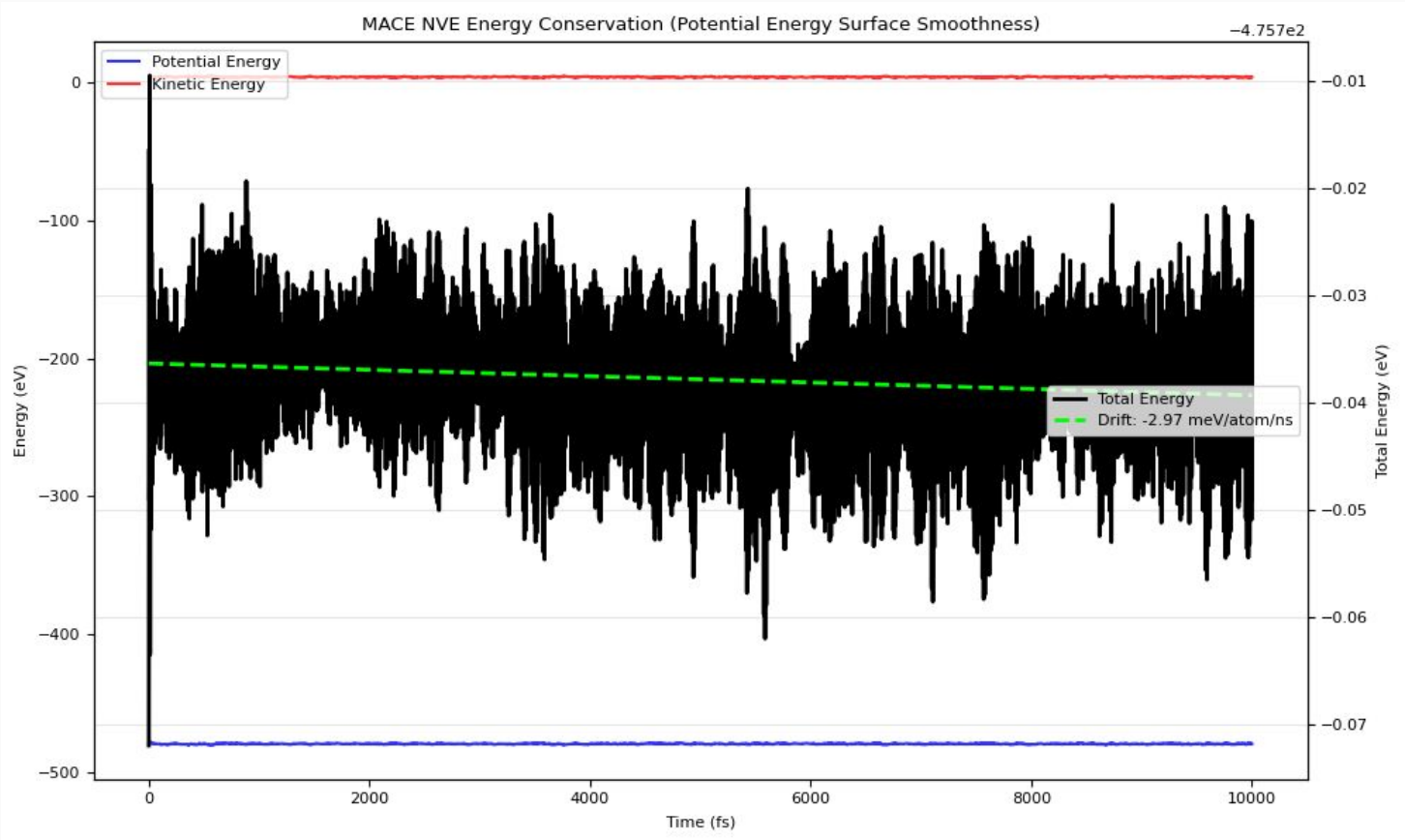
[Hydrogen (H)]
  Atom # 0: Max distortion 3.33 sigma | 81.7% inside 2-sigma limit
  Atom #28: Max distortion 2.16 sigma | 99.6% inside 2-sigma limit
  Atom #55: Max distortion 3.75 sigma | 75.7% inside 2-sigma limit
[Carbon (C)]
  Atom # 0: Max distortion 3.77 sigma | 92.9% inside 2-sigma limit
  Atom # 8: Max distortion 3.27 sigma | 88.8% inside 2-sigma limit
  Atom #15: Max distortion 3.55 sigma | 93.0% inside 2-sigma limit
[Nitrogen (N)]
  Atom # 0: Max distortion 4.02 sigma | 84.0% inside 2-sigma limit
  Atom # 4: Max distortion 4.65 sigma | 79.5% inside 2-sigma limit
  Atom # 7: Max distortion 3.03 sigma | 90.9% inside 2-sigma limit
[Tin (Sn)]
  Atom # 0: Max distortion 2.74 sigma | 88.8% inside 2-sigma limit
  Atom # 2: Max distortion 3.99 sigma | 80.9% inside 2-sigma limit
  Atom # 3: Max distortion 2.97 sigma | 93.2% inside 2-sigma limit
[Iodine (I)]
  Atom # 0: Max distortion 2.48 sigma | 97.2% inside 2-sigma limit
  Atom # 8: Max distortion 2.74 sigma | 98.7% inside 2-sigma limit
  Atom #15: Max distortion 3.36 sigma | 80.6% inside 2-sigma limit

MACE - Quality of the fit: agreement between models

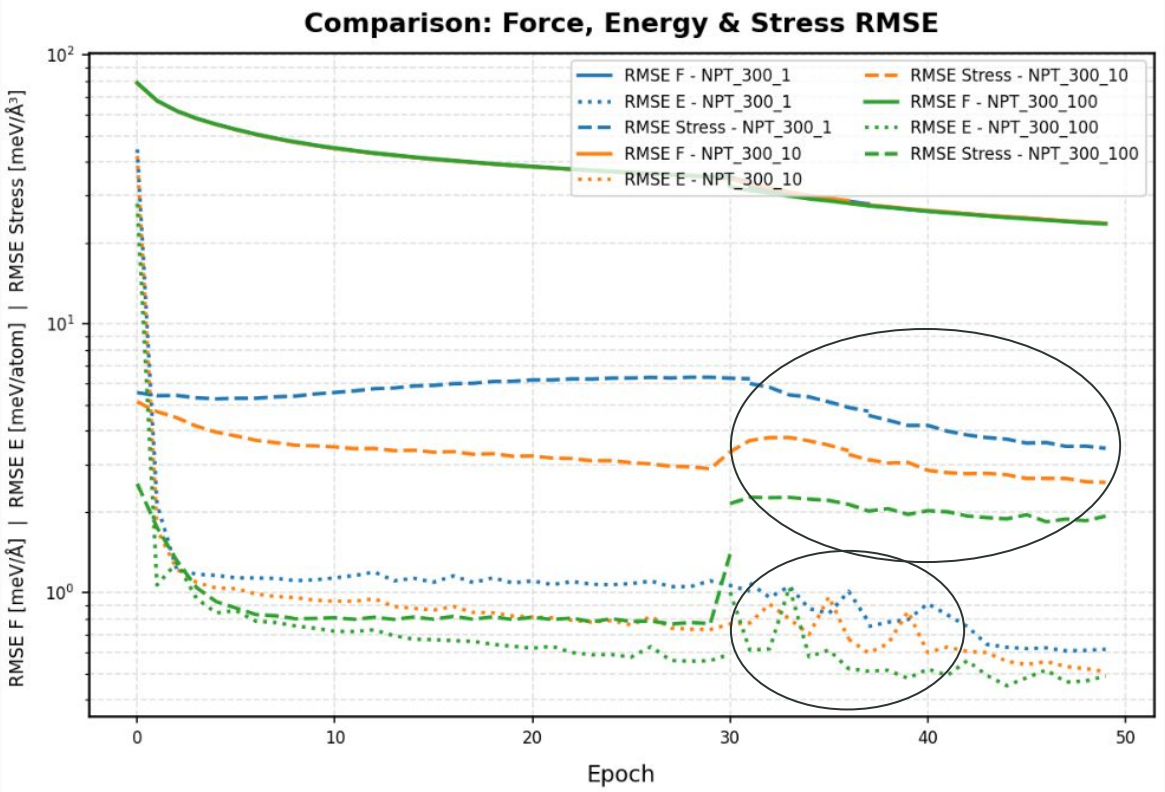
Taking the same model with different seeds that dictates the generation of pseudo-random values, we can see whether a model randomly guesses a frame (hence the model never saw it) or makes an educated guess by seeing if the different models agree on physical predictions



MACE - Quality of the fit: NVE analysis



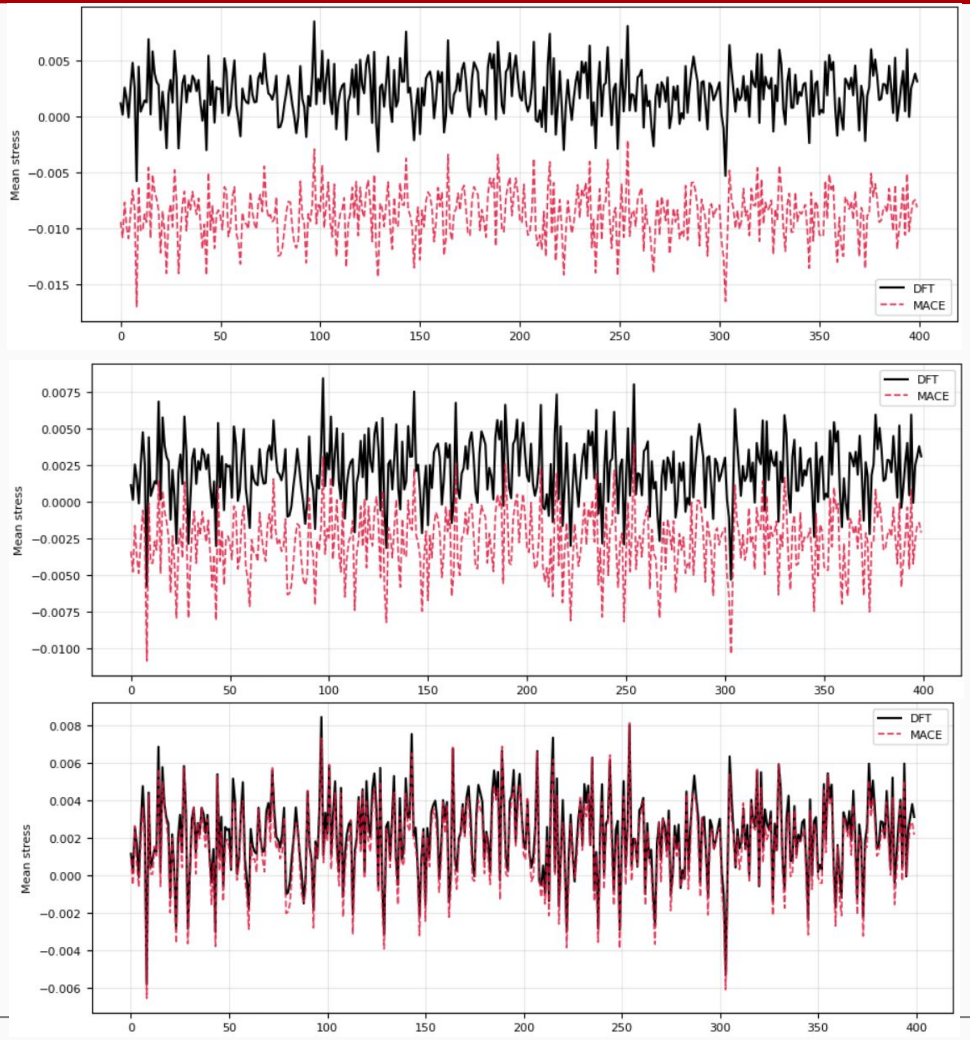
3 stress_weight: 1, 10 , 100



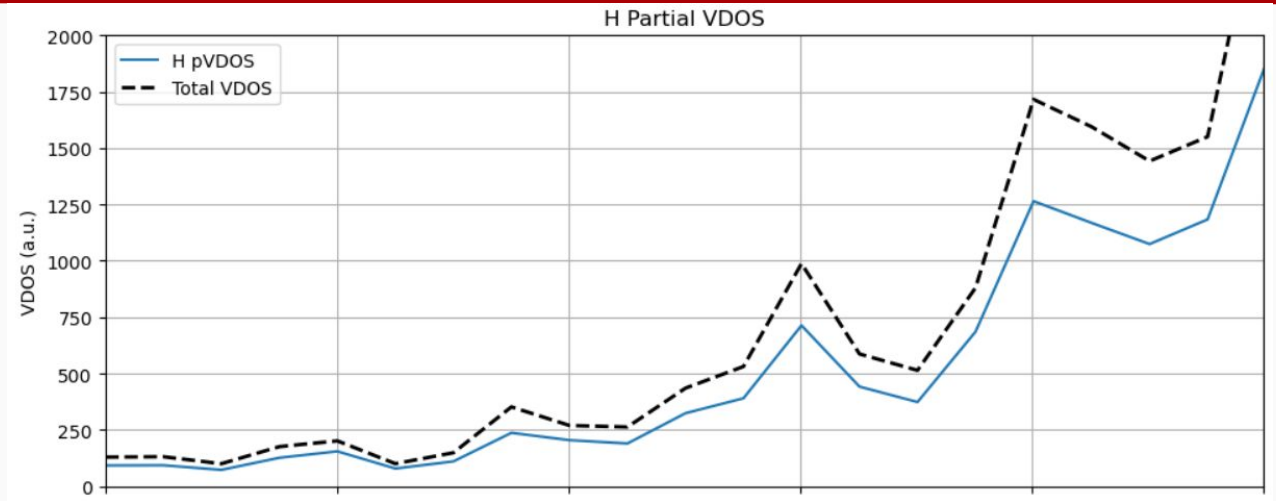
1

10

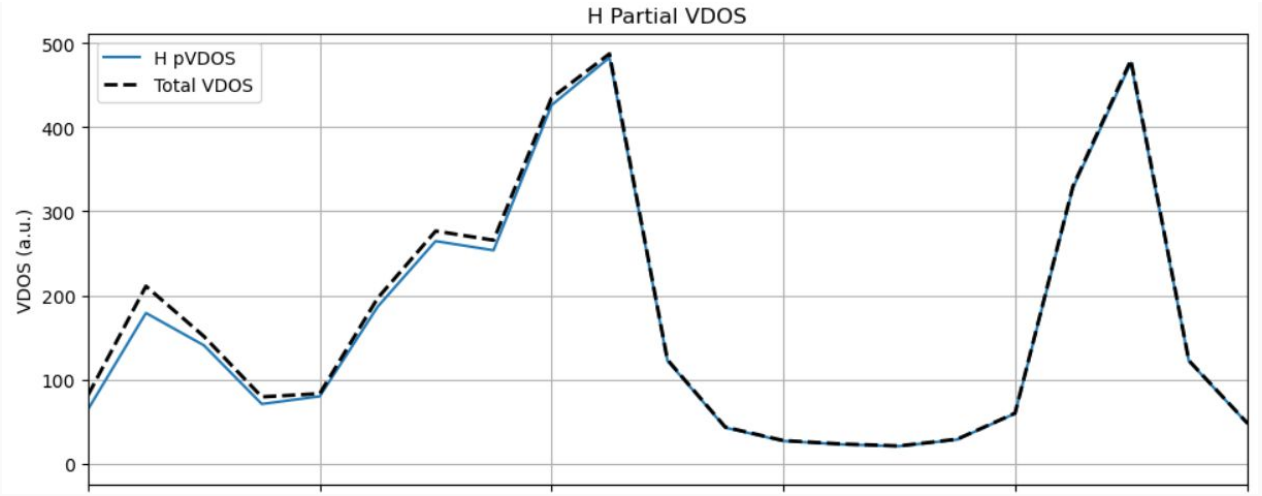
100


Mean stress
DFT
MACE
0.005
0.000
−0.005
−0.010
−0.015
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
0
50
100
150
200
250
300
350
400

DFT


H Partial VDOS
H pVDOS
Total VDOS
VDOS (a.u.)
2000
1750
1500
1250
1000
750
500
250
0

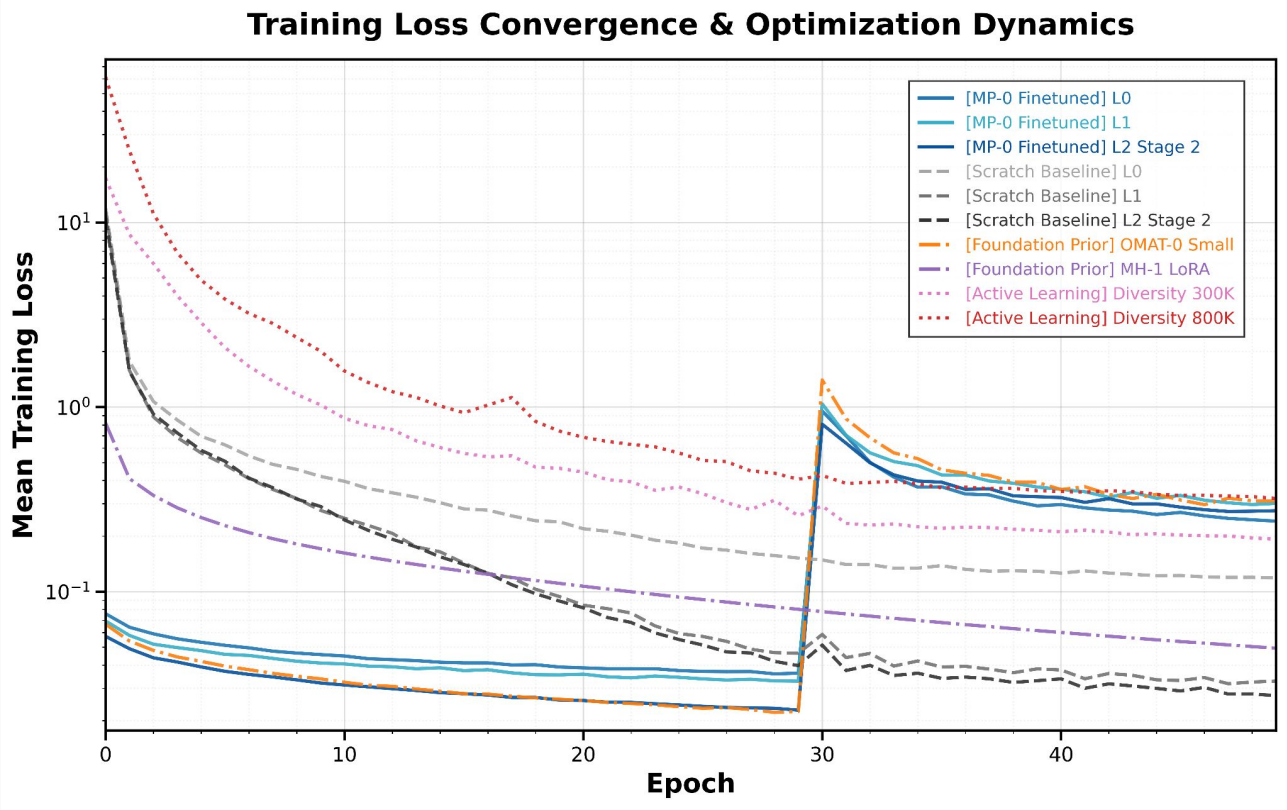
MACE


H Partial VDOS
H pVDOS
Total VDOS
VDOS (a.u.)
500
400
300
200
100
0

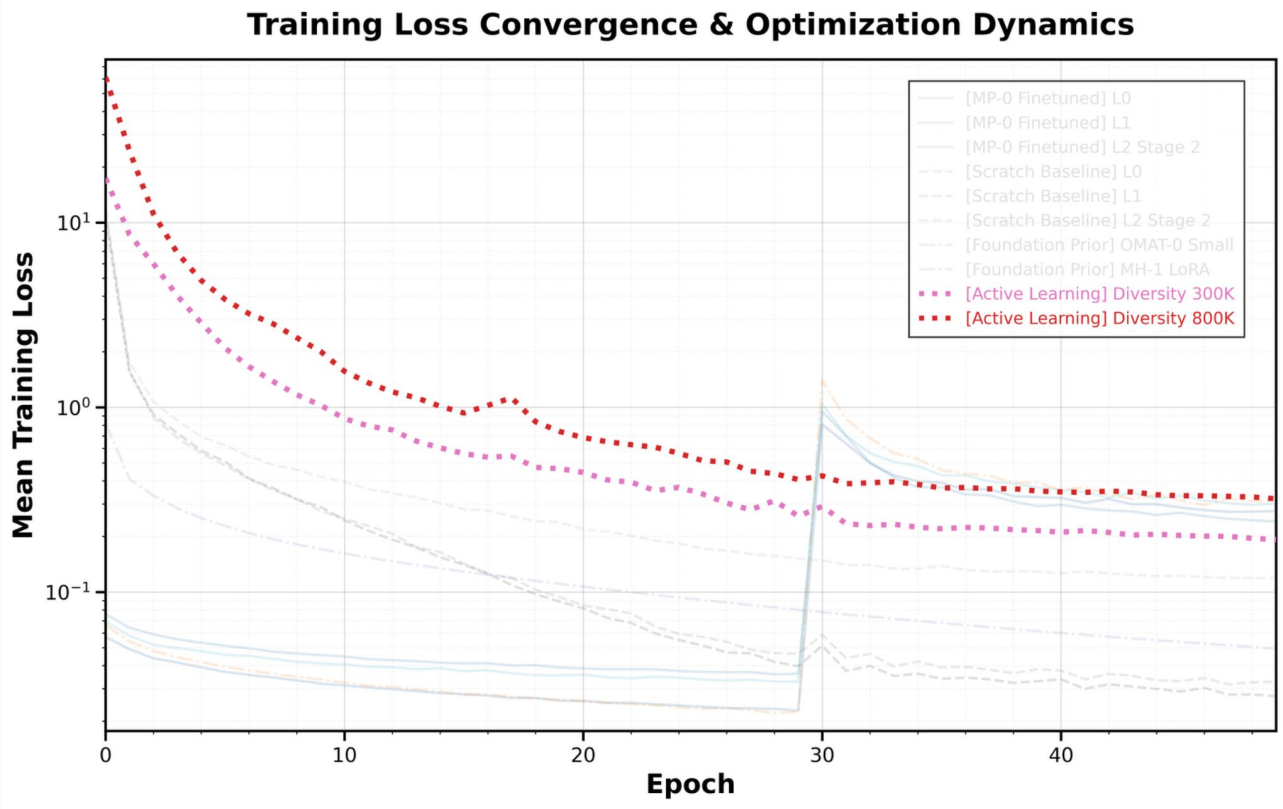
Stress-Testing the MACE Potentials: Benchmarking & Validation

Model Family	Specific Variants	Key Architectural Trait	Physical Motivation / Hypothesis
From-Scratch Baseline	`scratch_L0`, `scratch_L1`, `scratch_L2`	Random weight initialization; no prior chemical knowledge.	Establishes the baseline cost and data hunger of learning complex 5-element hybrid perovskites directly from ab initio targets.
MP-0 Foundation (Fine-tuned)	`finetuned_L0`, `finetuned_L1`, `finetuned_L2`	Pre-trained on Materials Project; scaled angular resolutions (Lmax=0,1,2).	Evaluates whether pre-trained crystalline inorganic representations transfer smoothly to hybrid organic-inorganic lattices.
Domain Specialists & Multi-Heads	`finetuned_OMAT_small`, `finetuned_MH1_LoRA`	OMAT-0 (solid-state specialist); MH-1 (universal multi-head adapted via LoRA).	Tests whether specialized solid-state priors outperform universal foundation models adapted via low-rank parameter-efficient fine-tuning (LoRA).
Out-of-Distribution Stress Sets	`diversity_300K`, `diversity_800K`	Active learning configurations restricted to 300K vs. high-temperature (800K) sampling.	Directly probes where potentials break down under thermal extrapolation beyond their training envelopes.

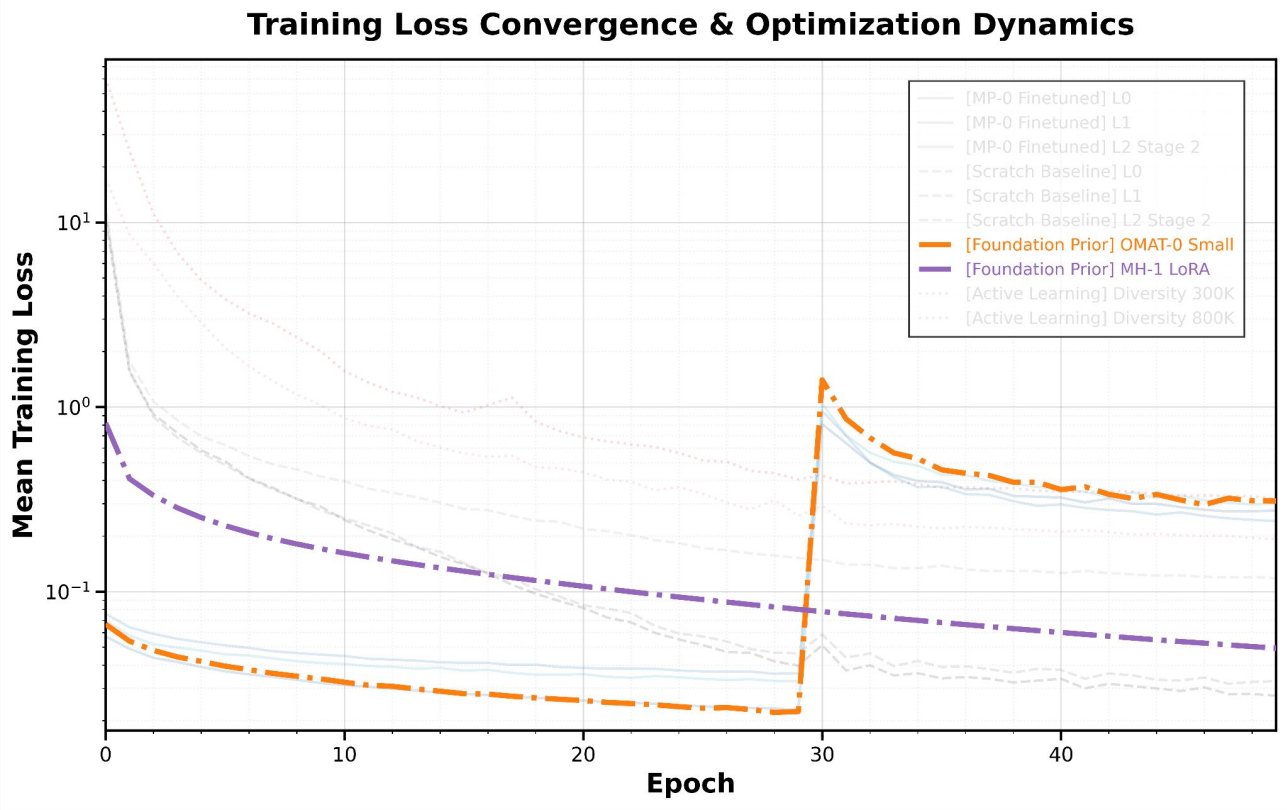
Training Dynamics & Optimization Behavior



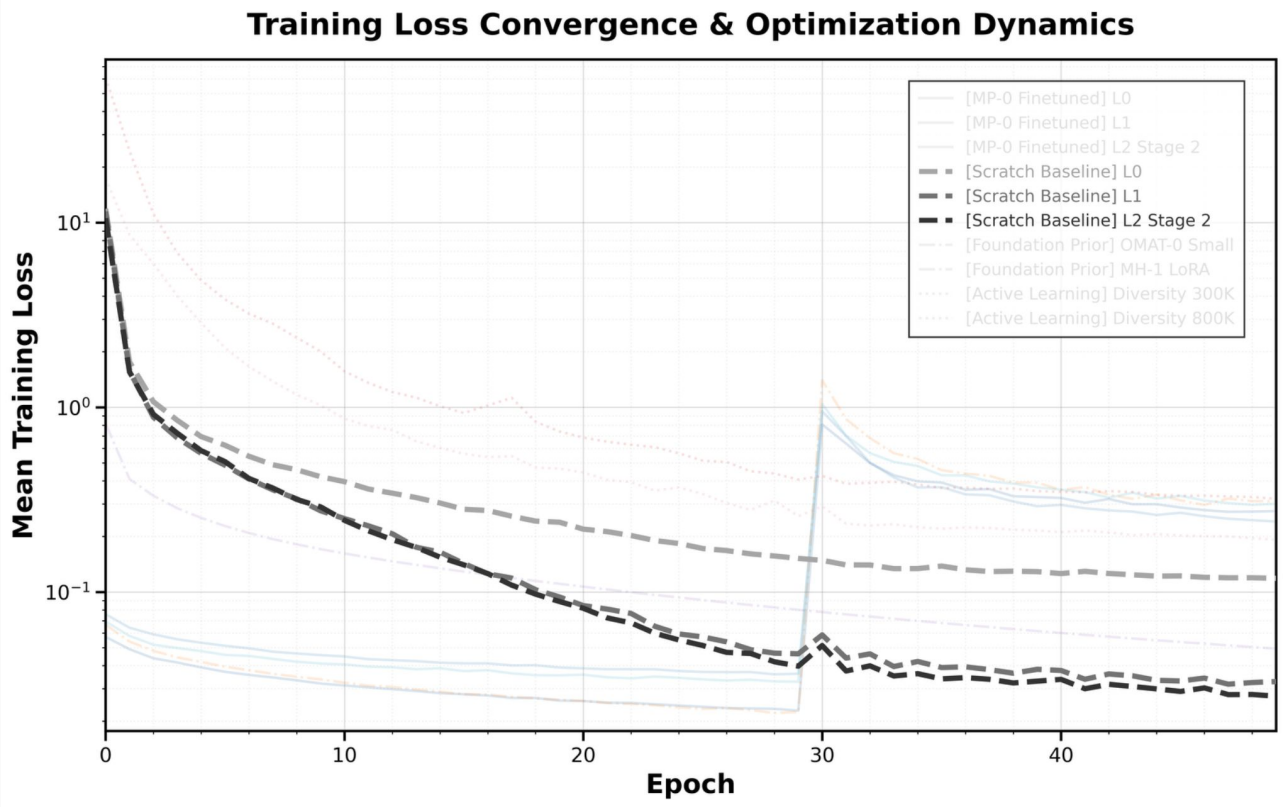
Training Dynamics & Optimization Behavior



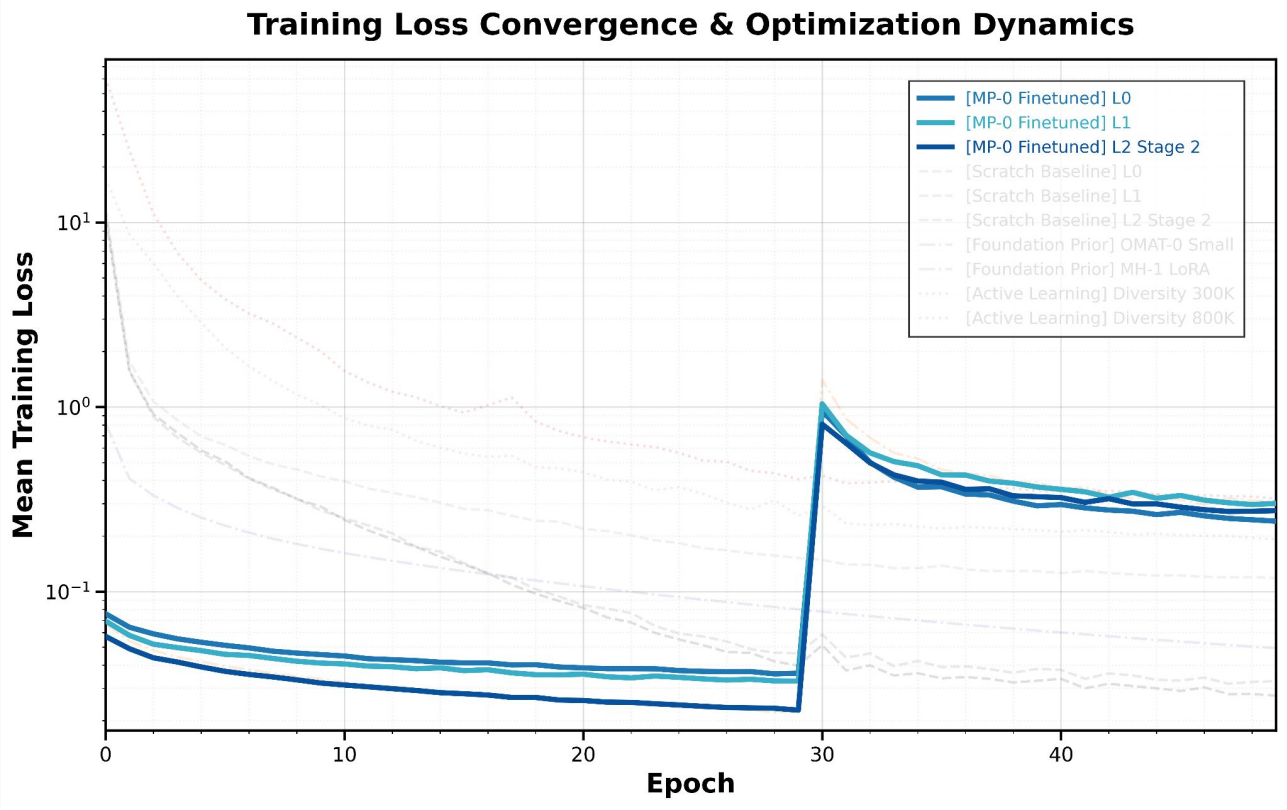
Training Dynamics & Optimization Behavior



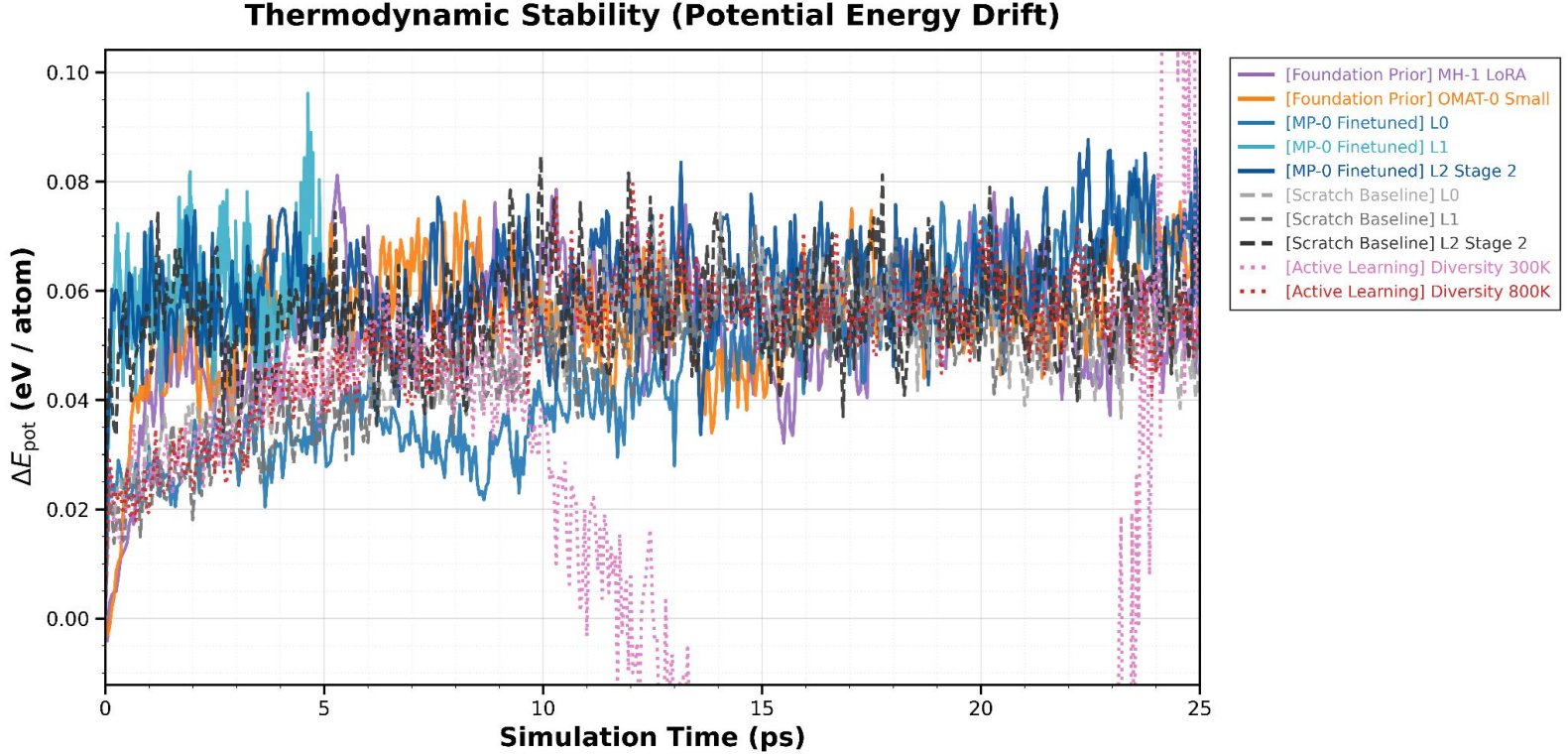
Training Dynamics & Optimization Behavior



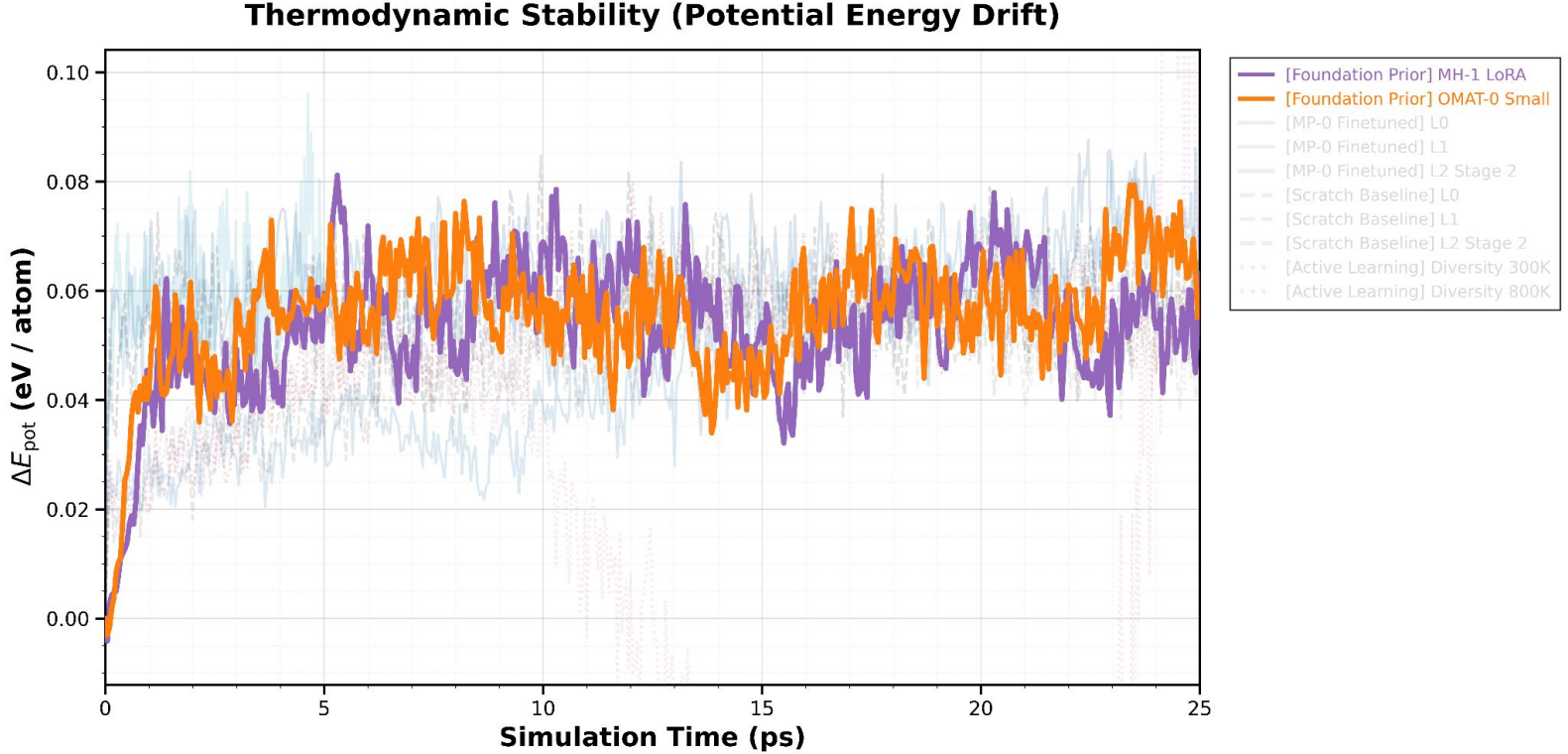
Training Dynamics & Optimization Behavior



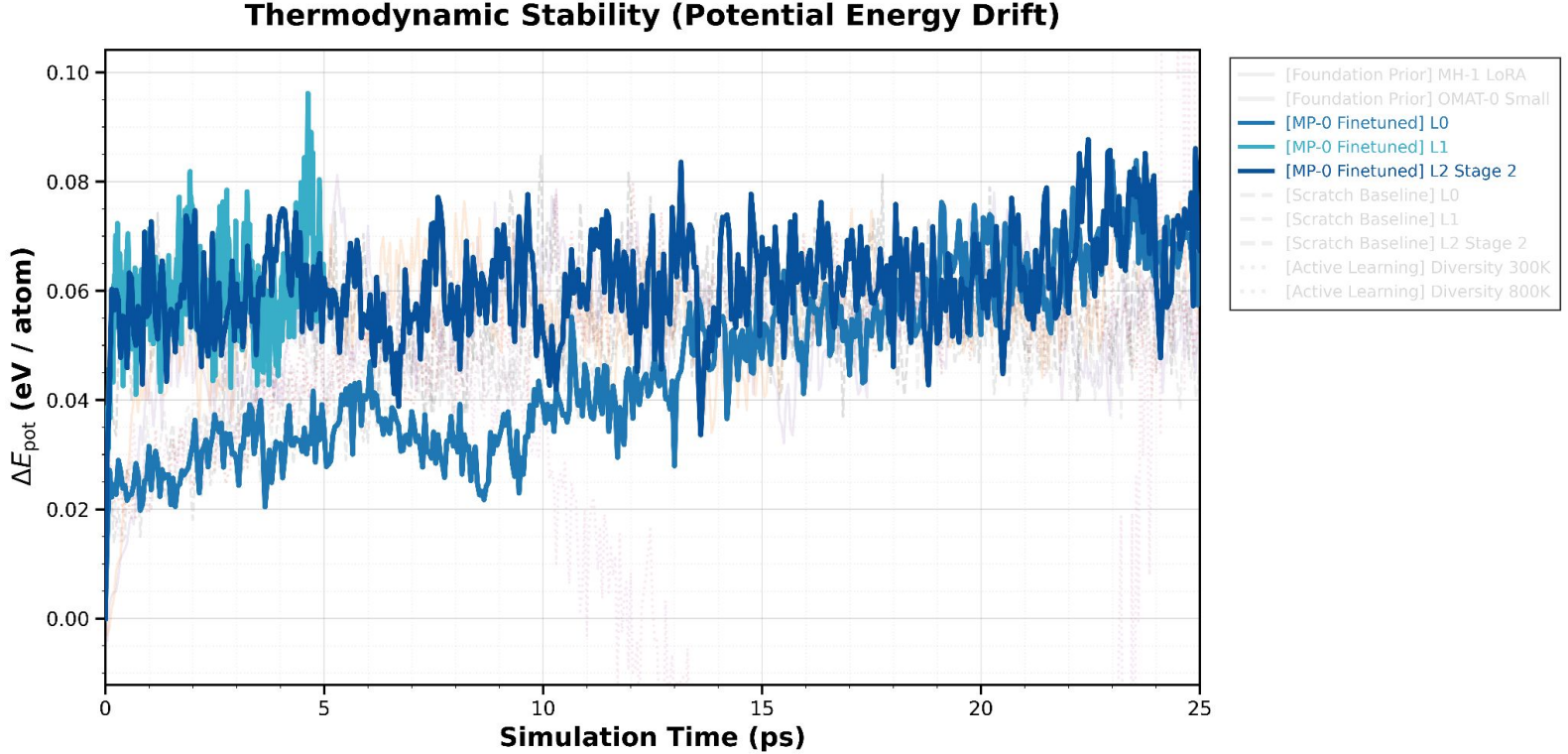
Thermodynamic Stability: Potential Energy Conservation



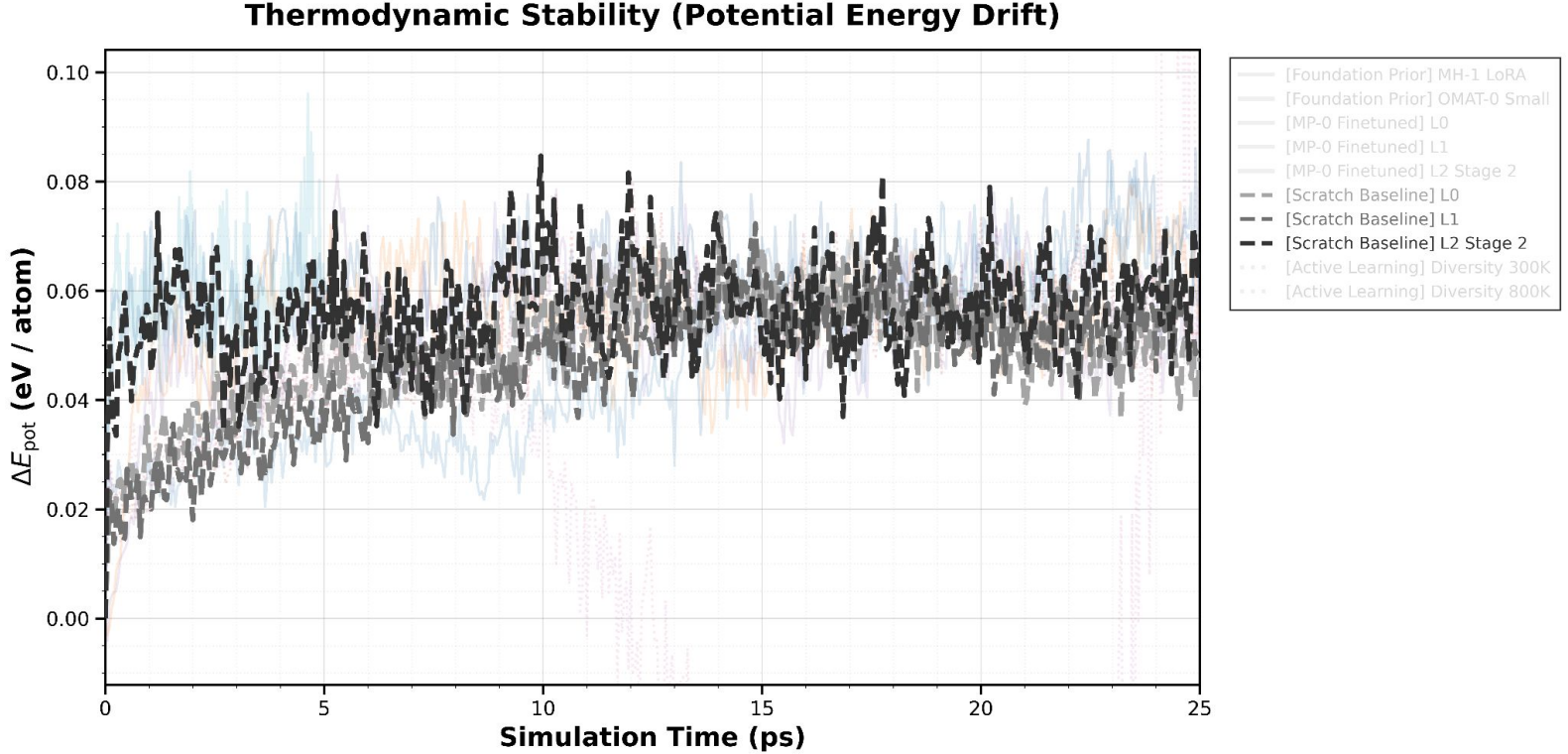
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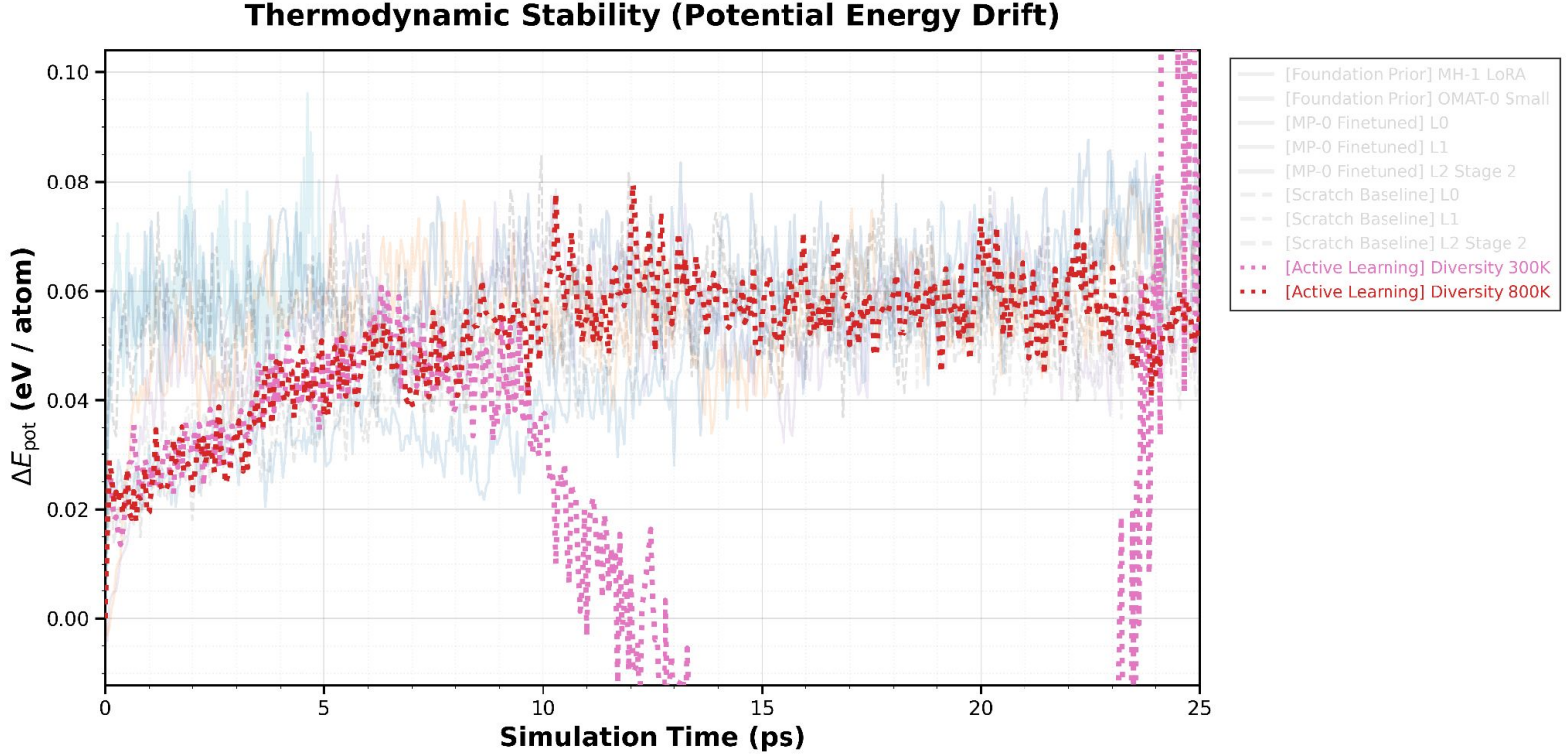
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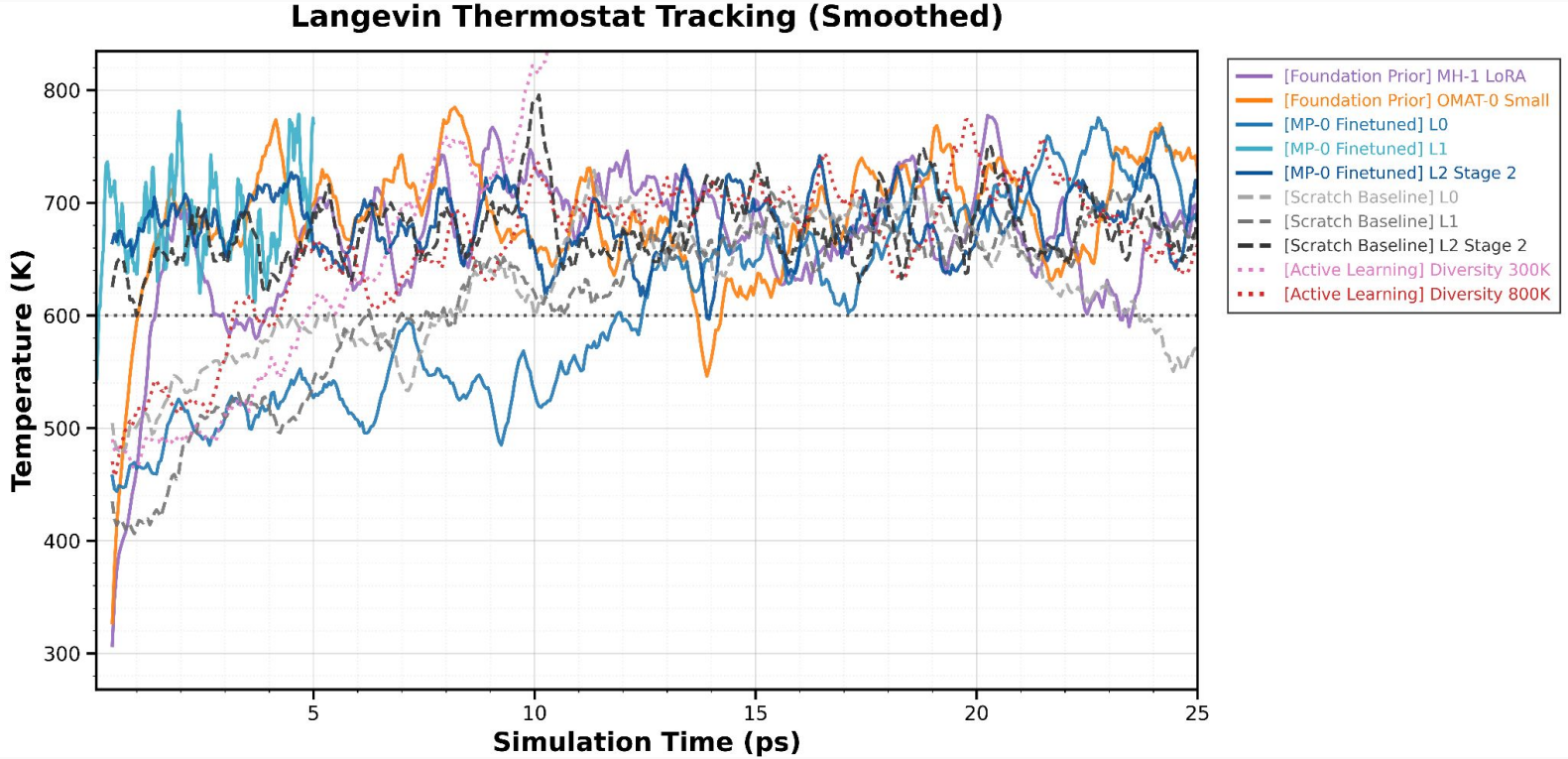
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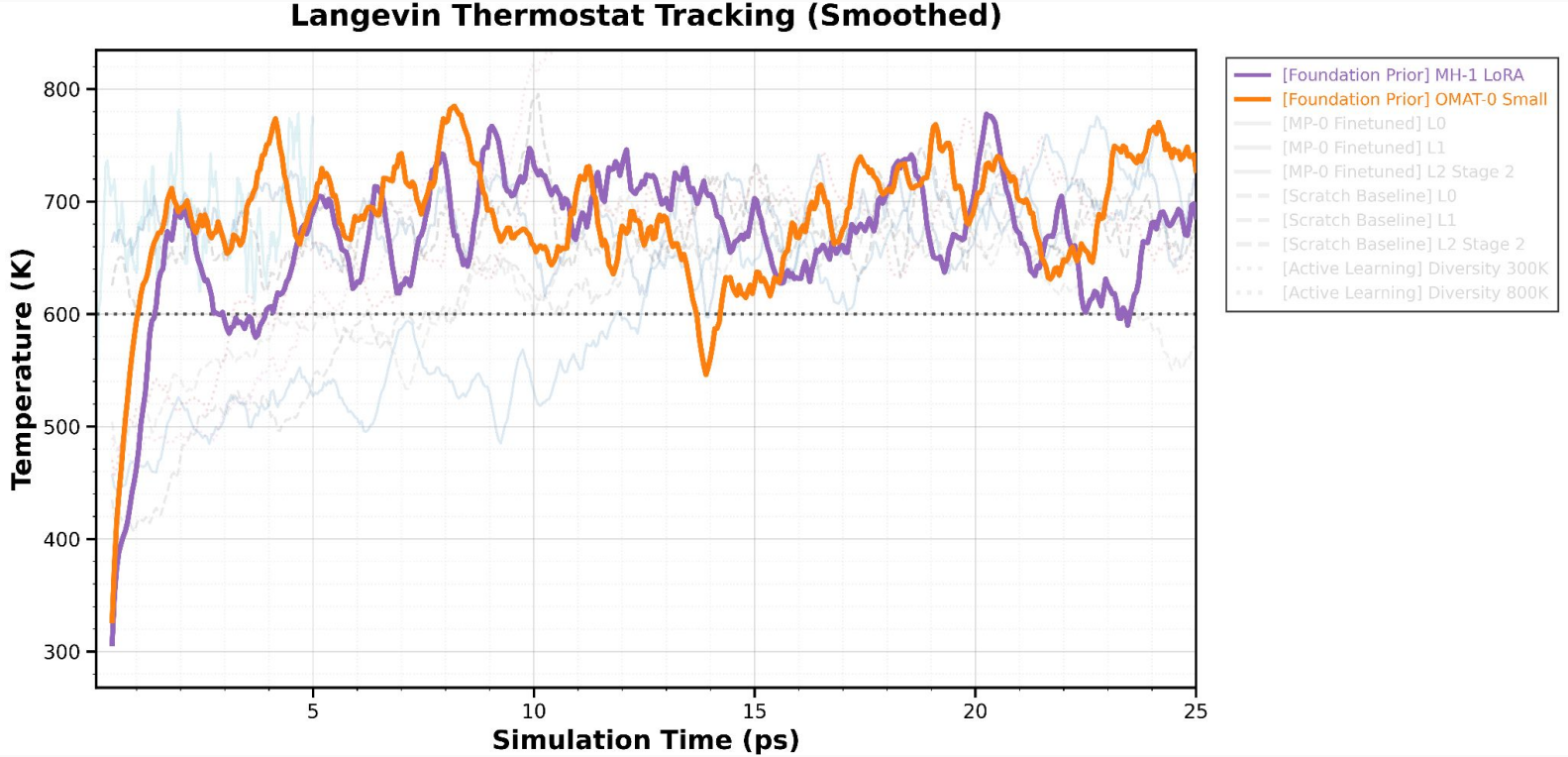
Thermodynamic Stability: Potential Energy Conservation



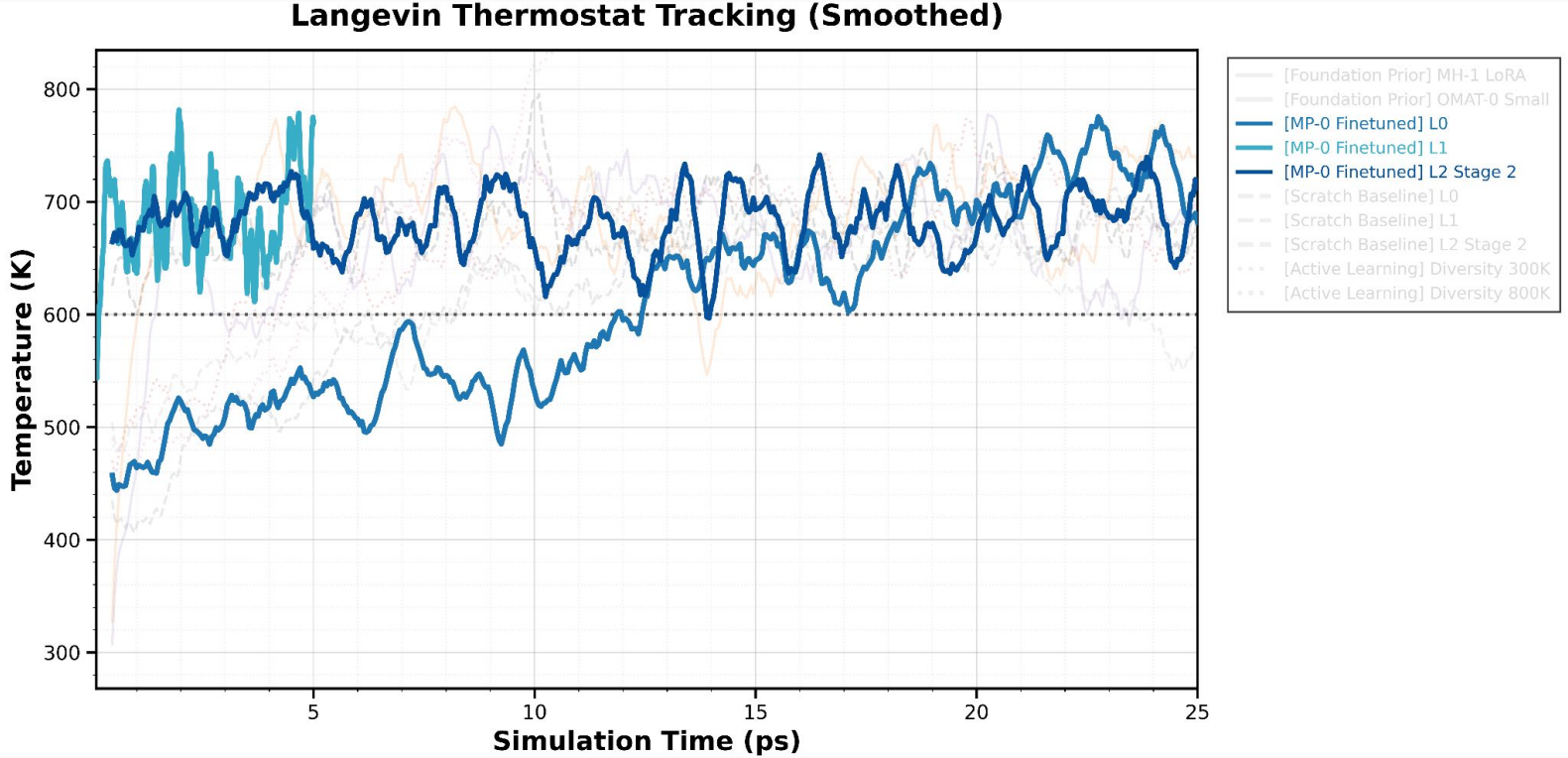
Temperature Tracking & Thermostat Response



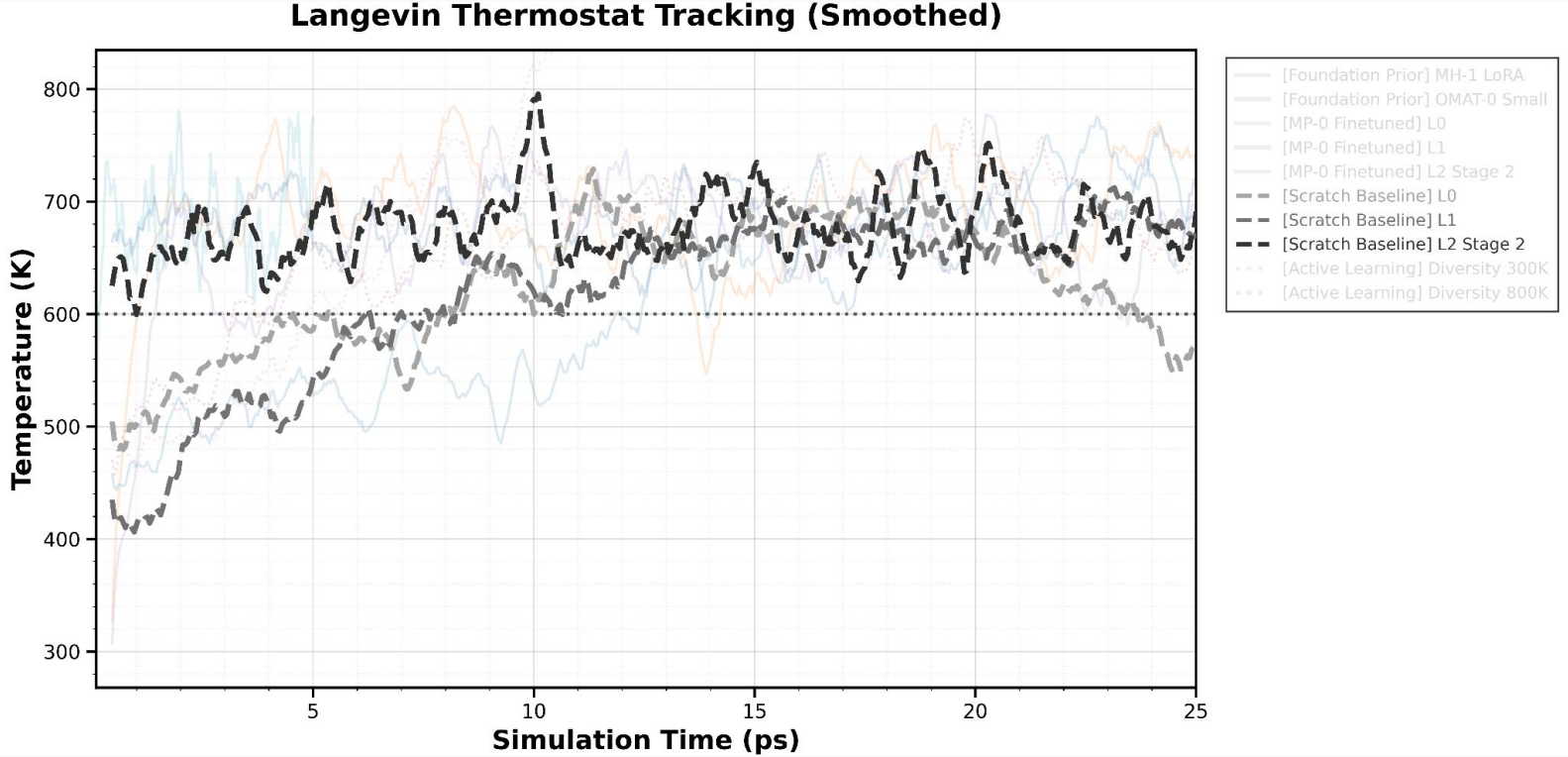
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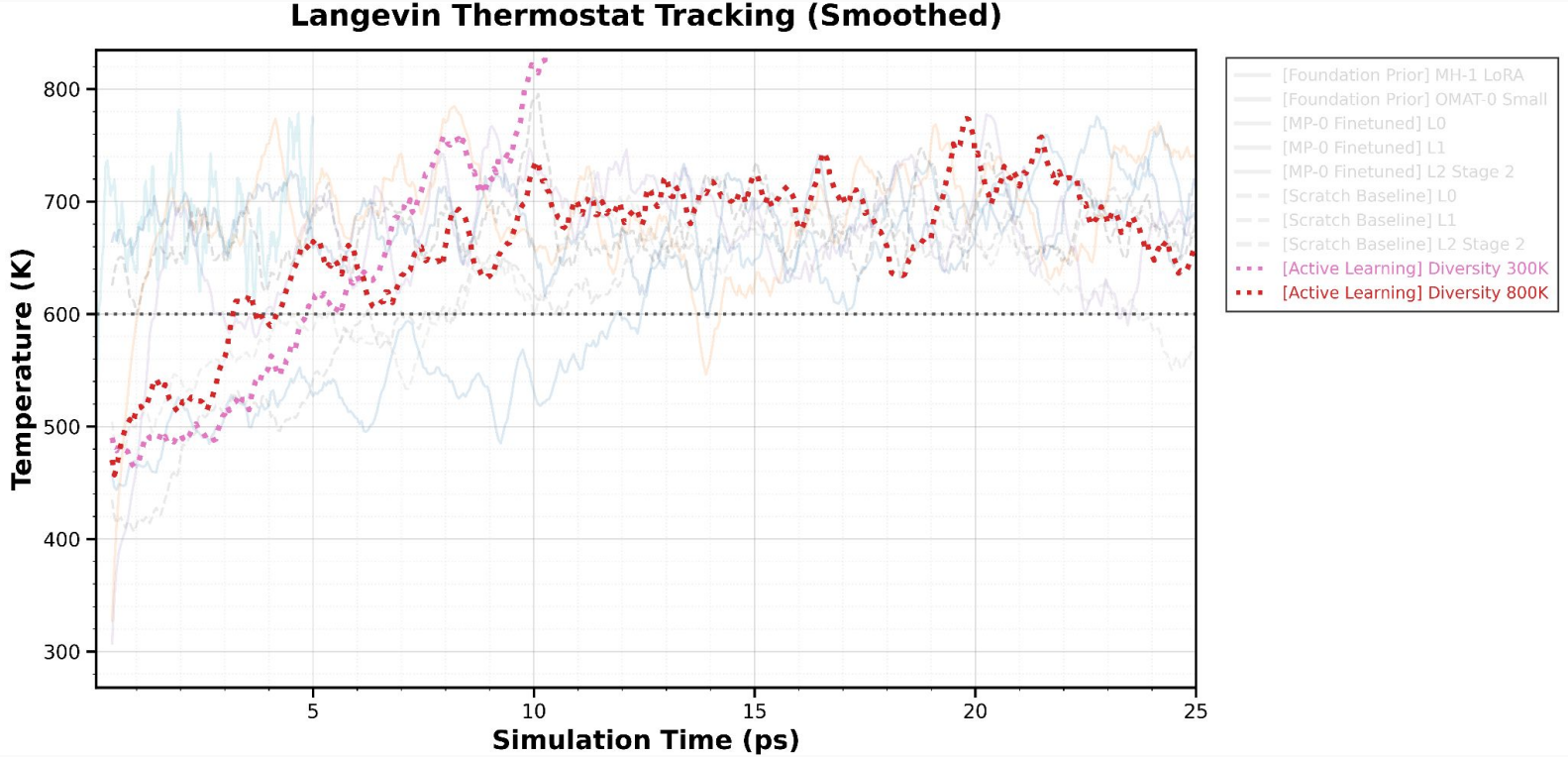
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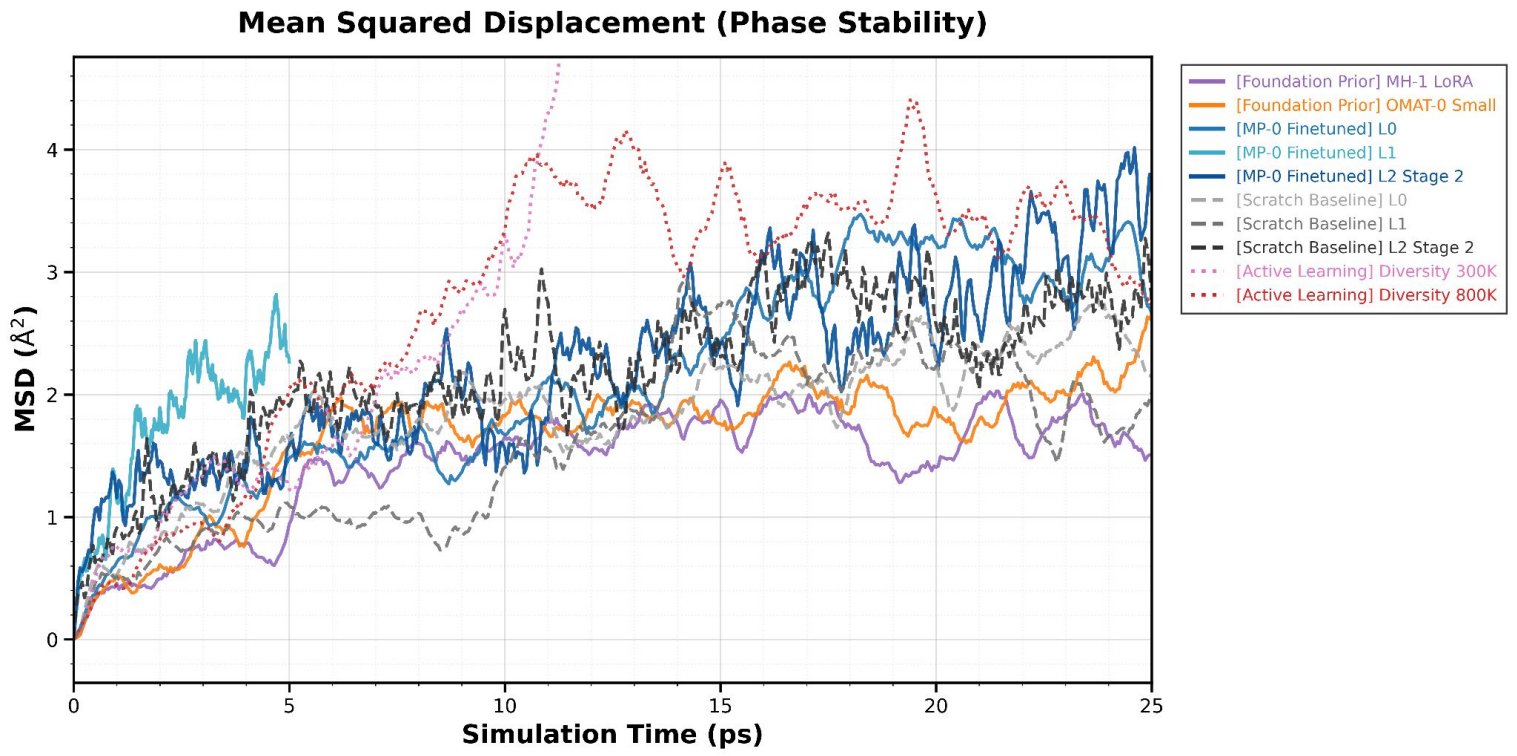
Temperature Tracking & Thermostat Response



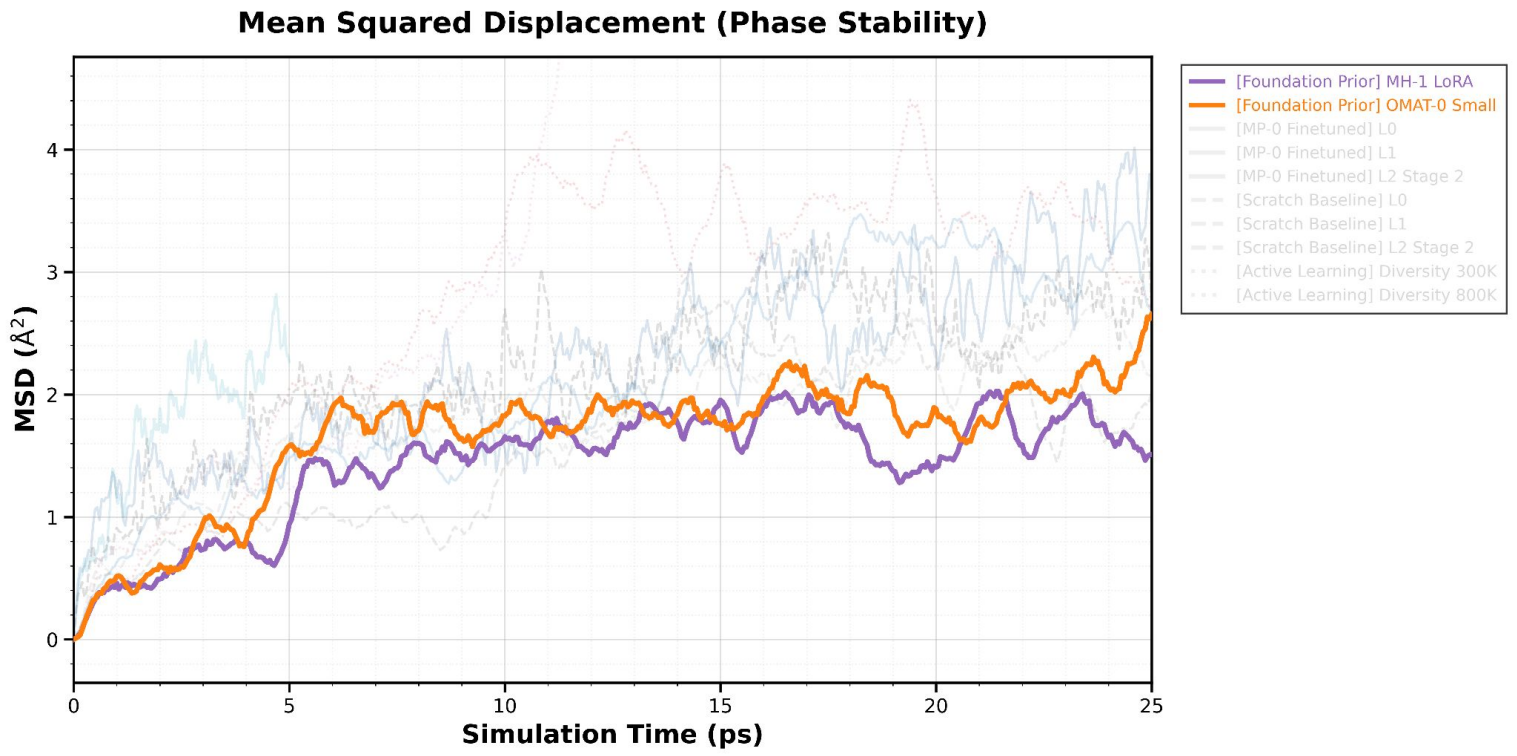
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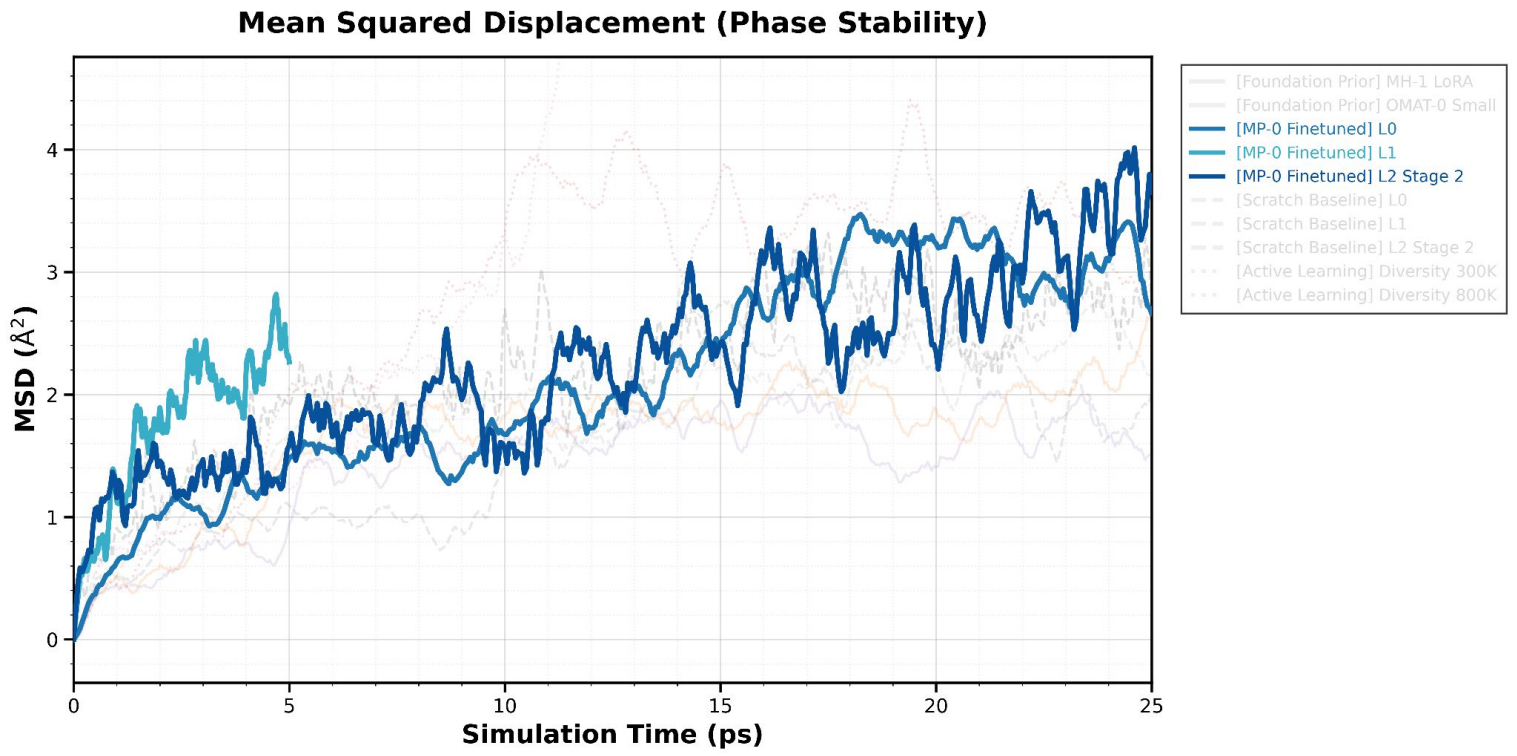
Phase Stability & Coordination Integrity



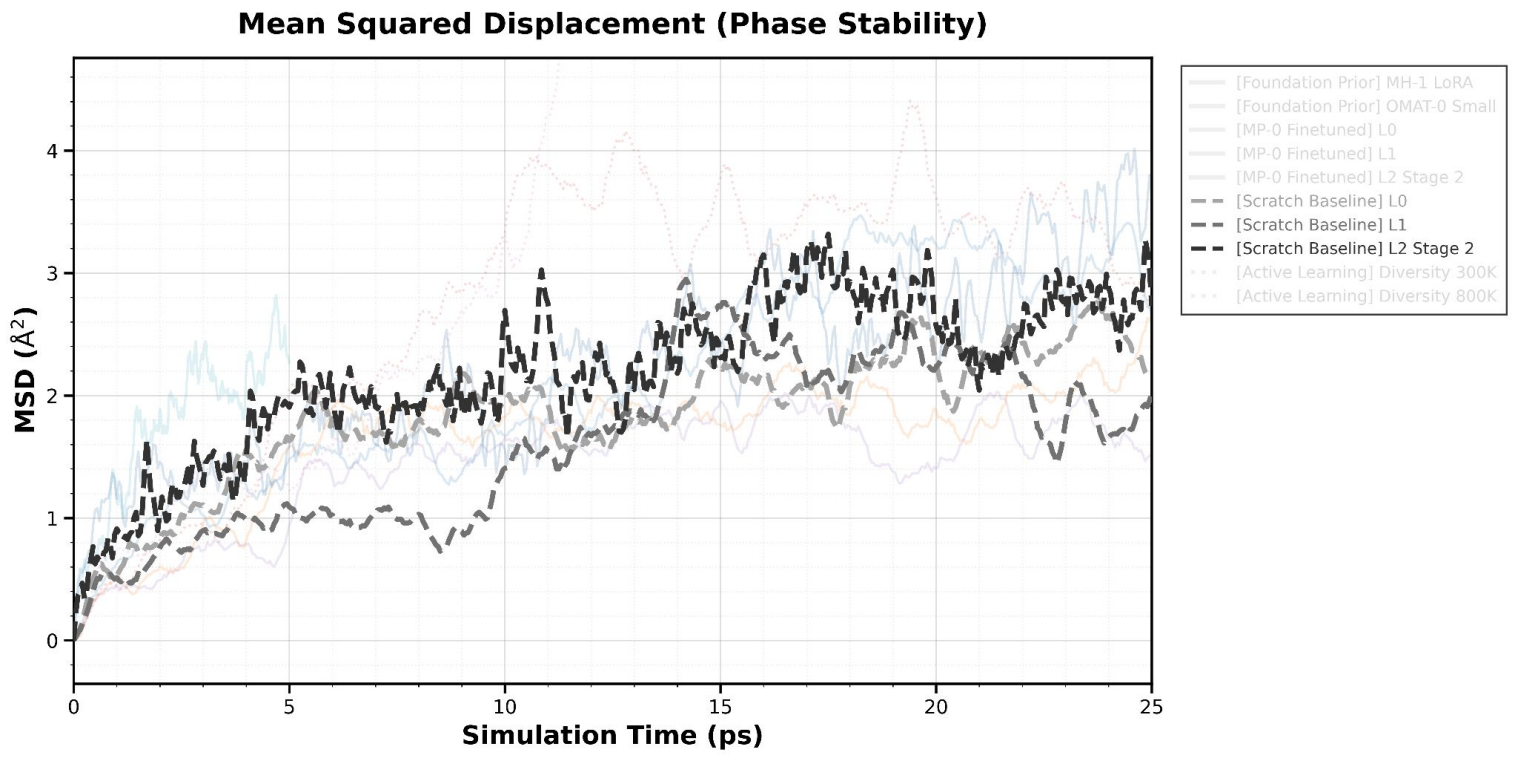
Phase Stability & Coordination Integrity



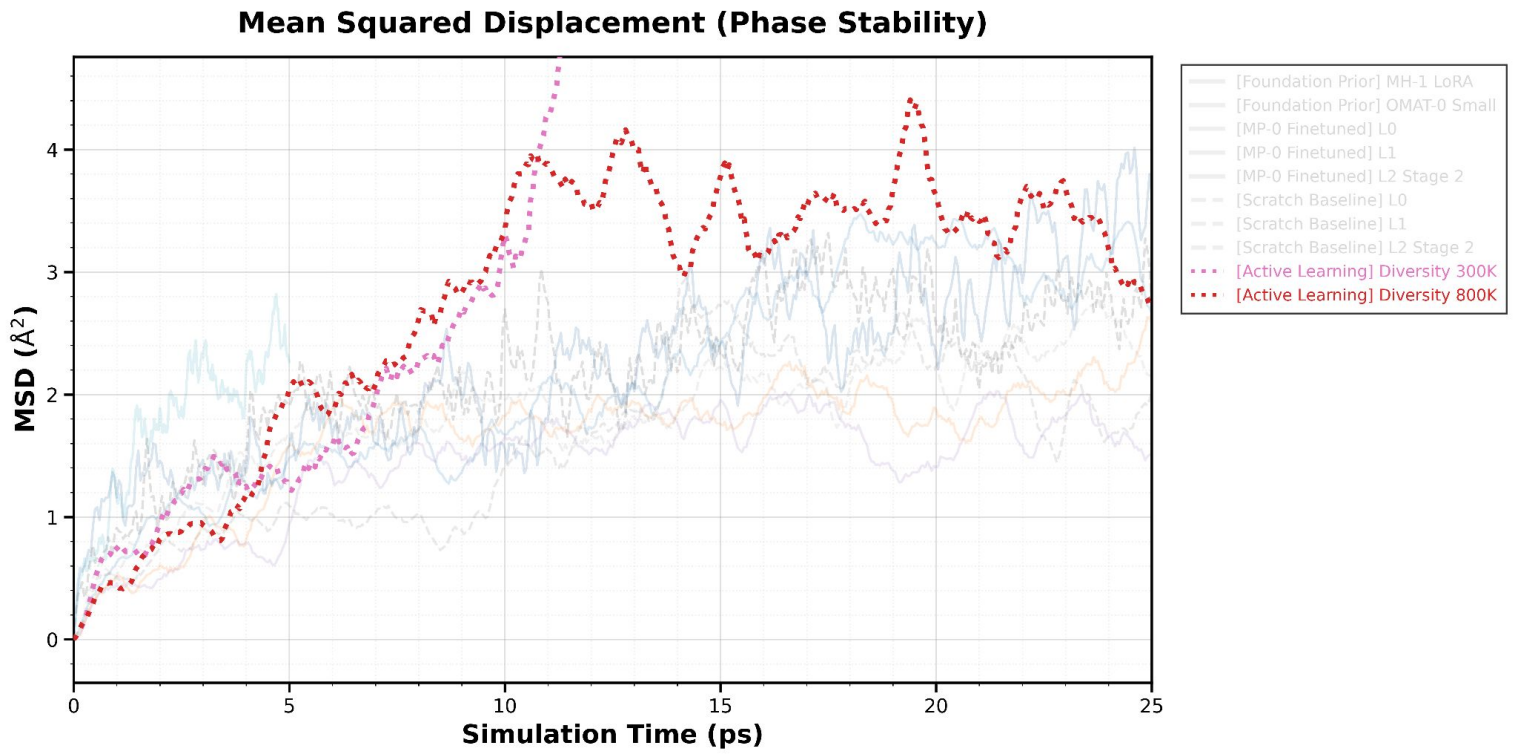
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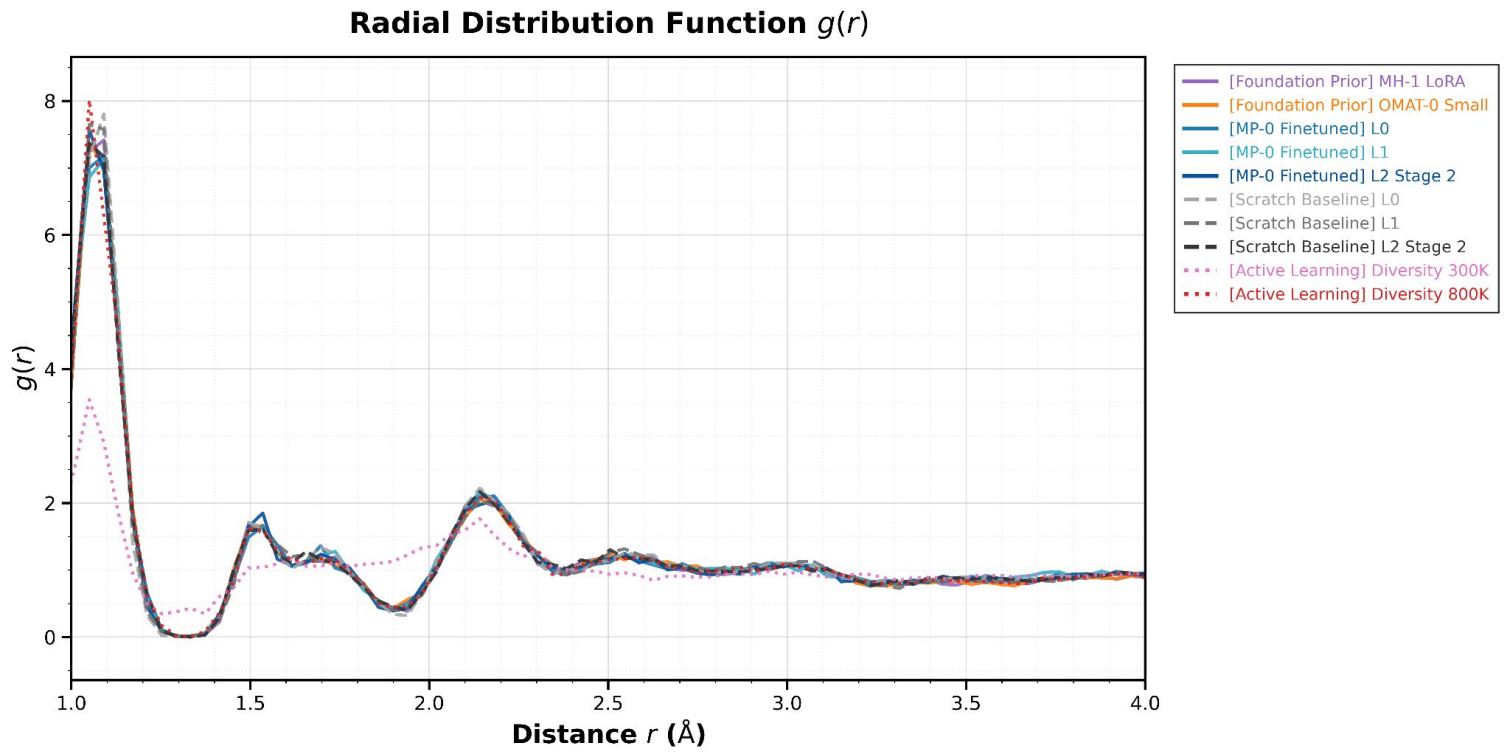
Phase Stability & Coordination Integrity



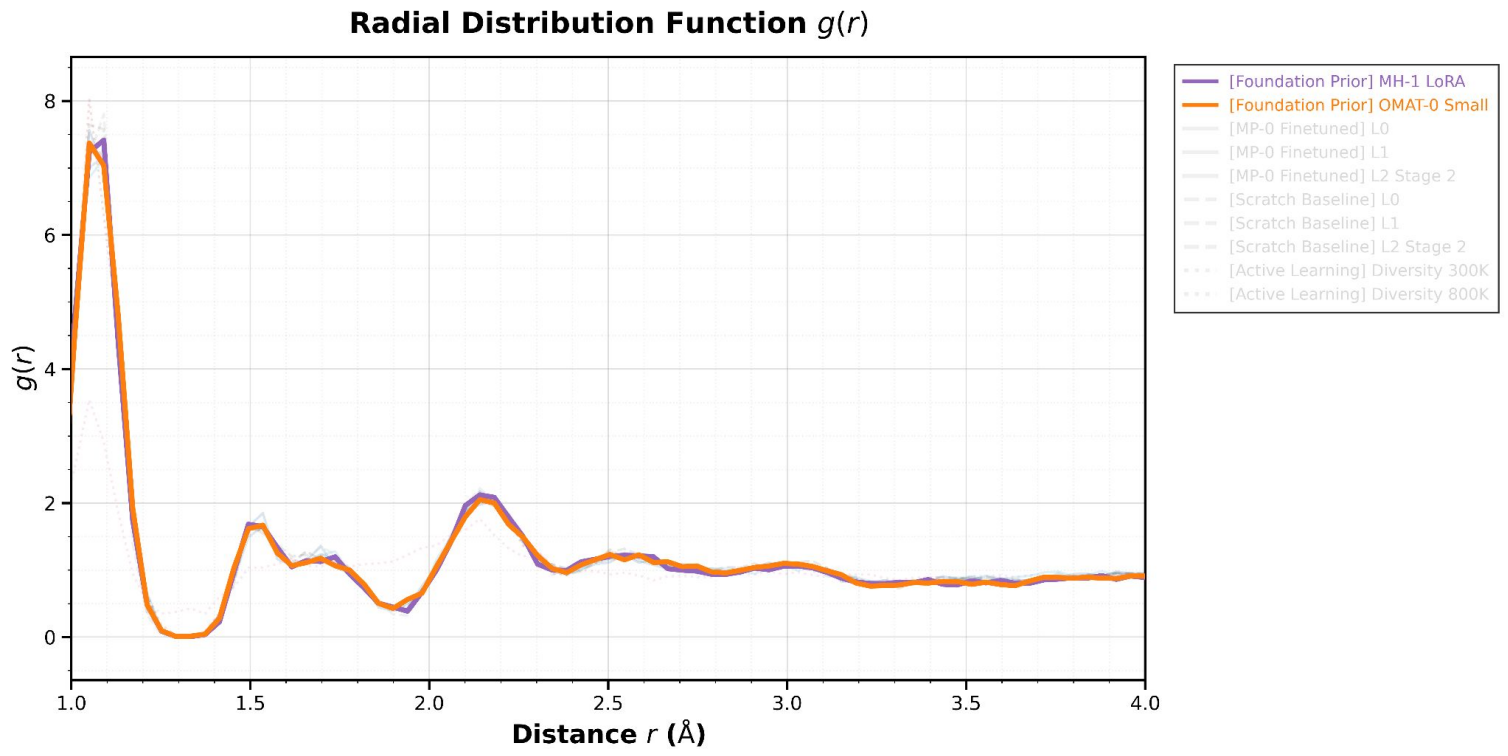
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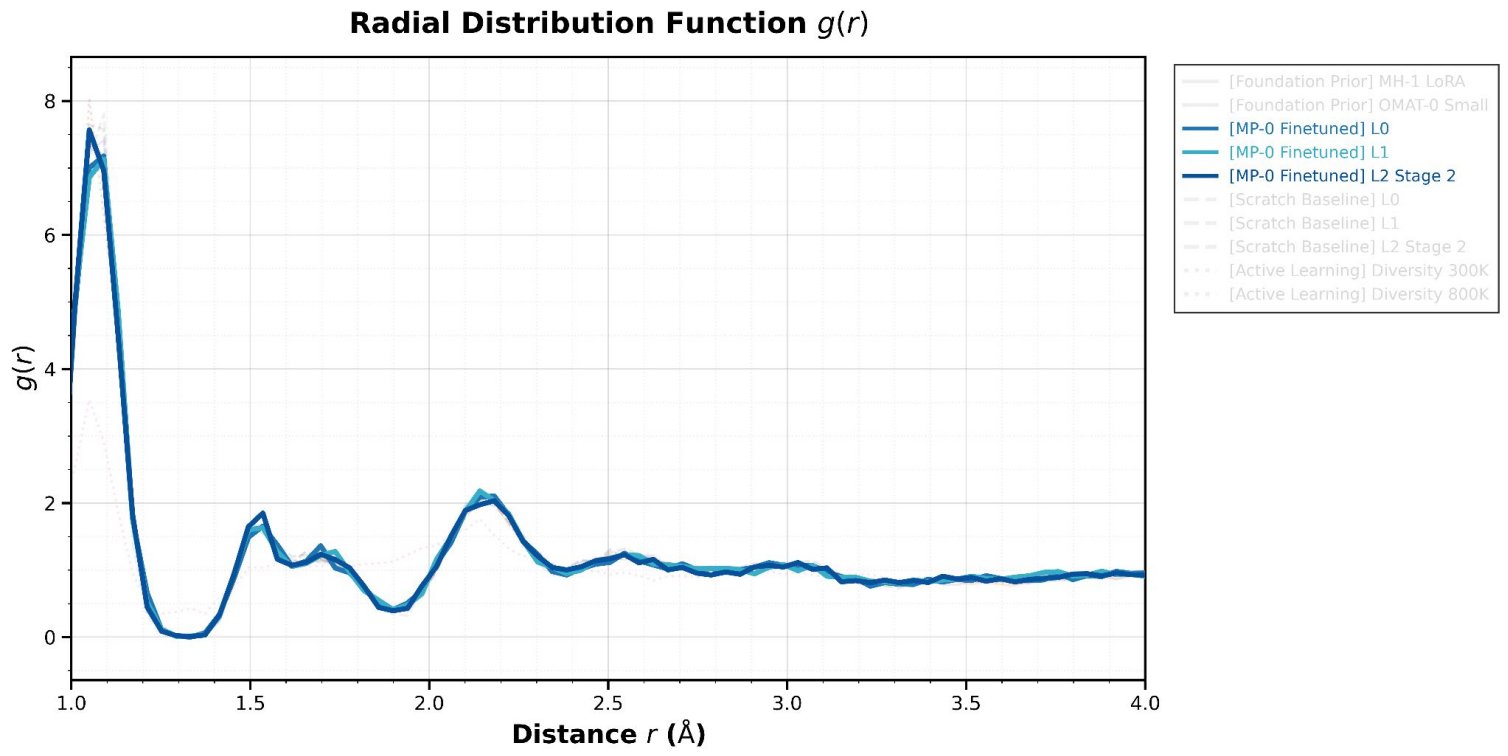
Systematic Benchmark: The Boundary Between Robustness and Failure



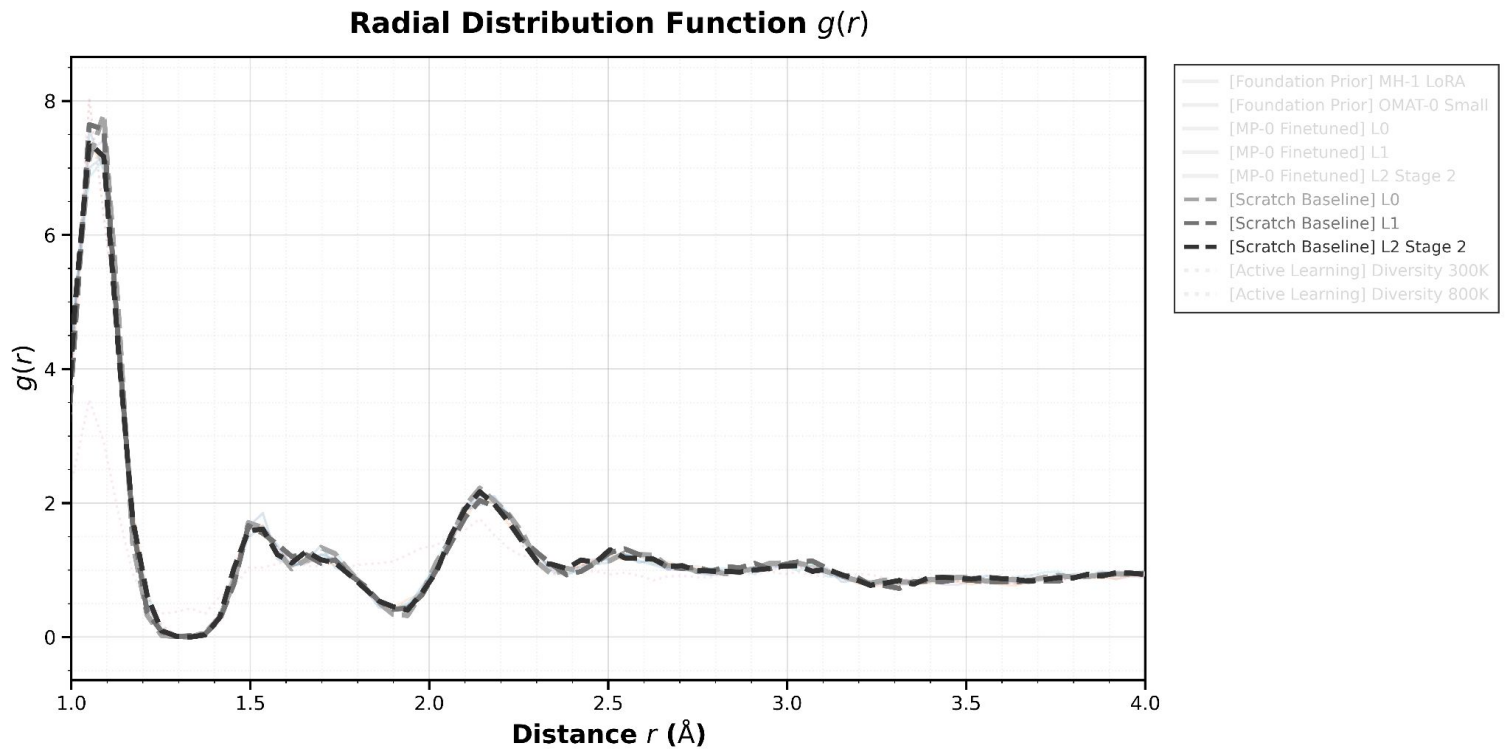
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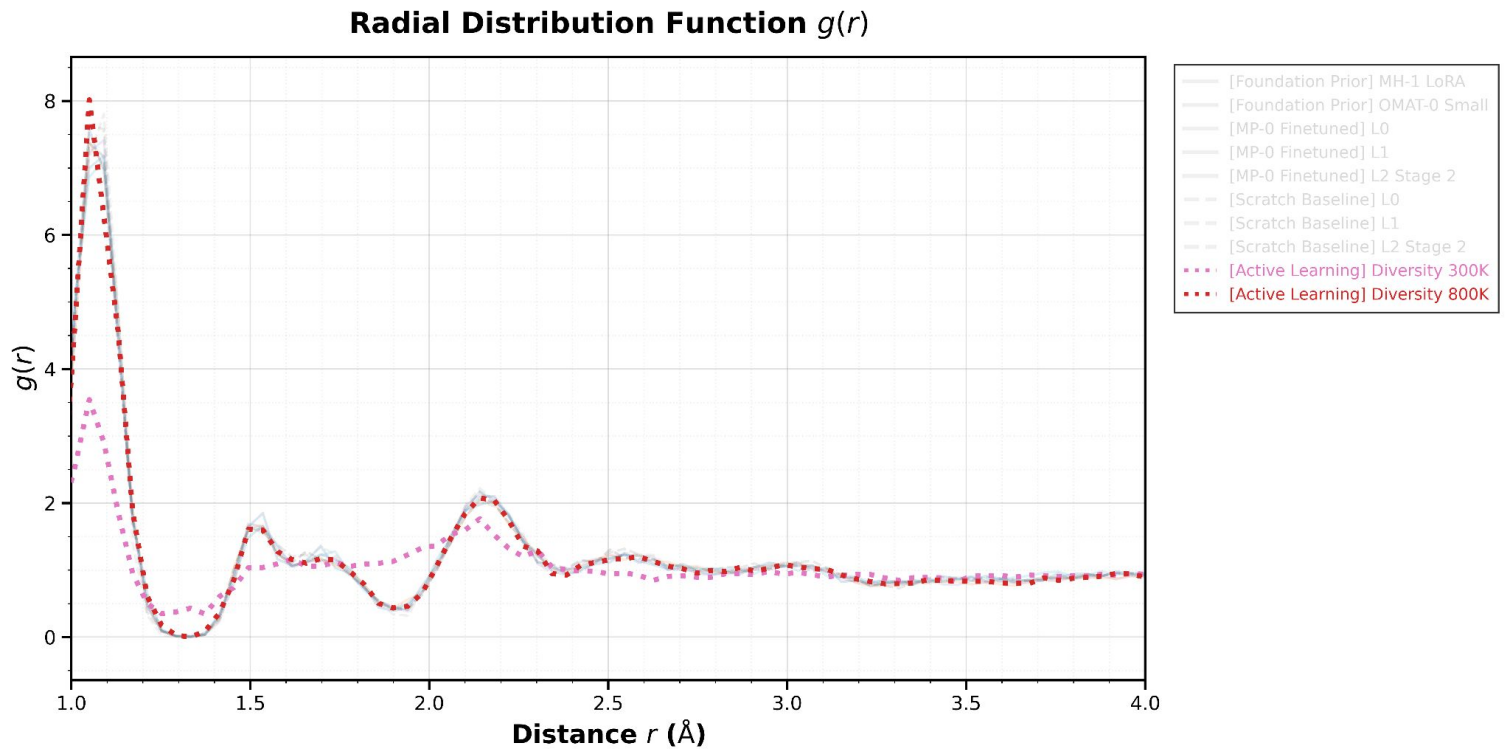
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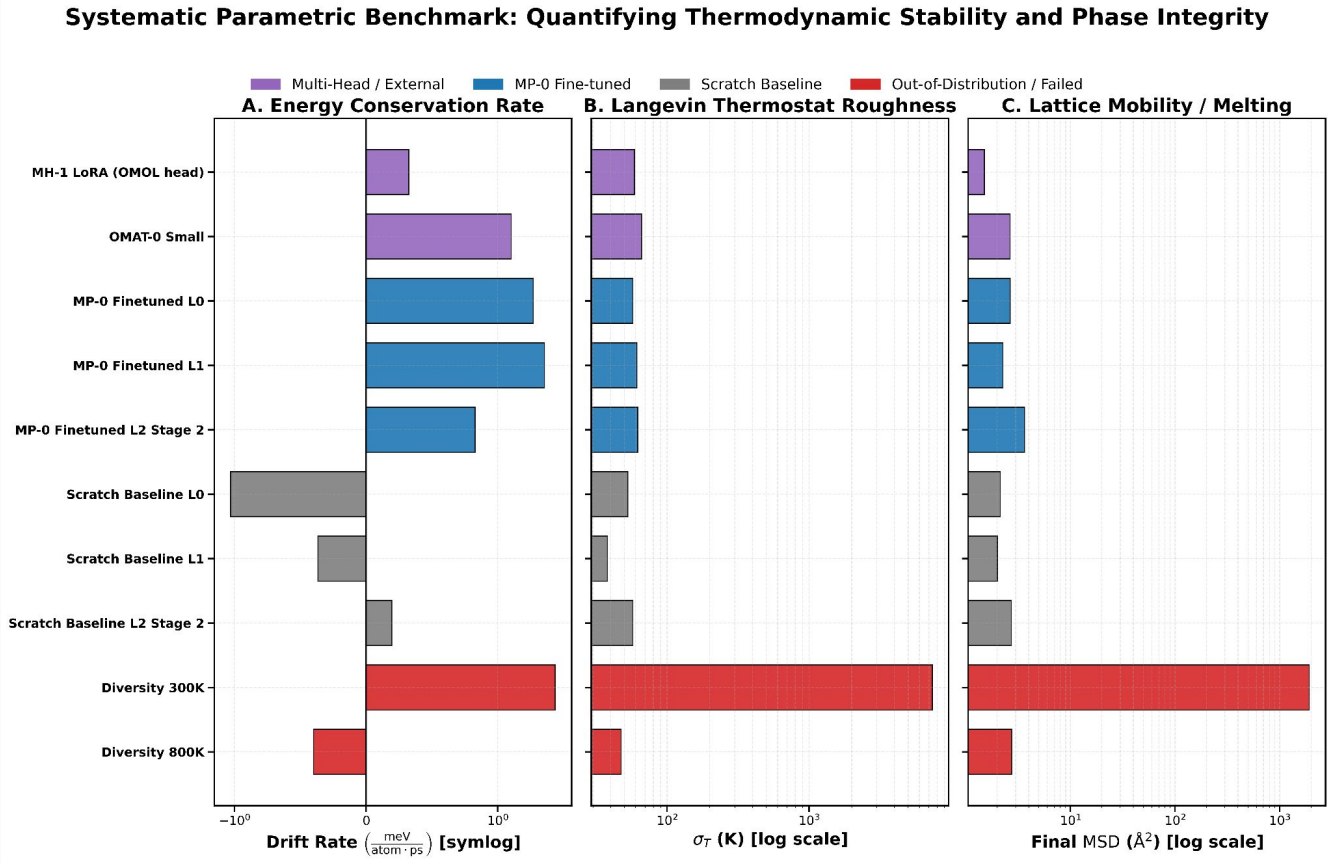


Systematic Benchmark: The Boundary Between Robustness and Failure



Systematic Benchmark: The Boundary Between Robustness and Failure




Systematic Parametric Benchmark: Quantifying Thermodynamic Stability and Phase Integrity
Multi-Head / External
MP-0 Fine-tuned
Scratch Baseline
Out-of-Distribution / Failed
A. Energy Conservation Rate
B. Langevin Thermostat Roughness
C. Lattice Mobility / Melting
MH-1 LoRA (OMOL head)
OMAT-0 Small
MP-0 Finetuned L0
MP-0 Finetuned L1
MP-0 Finetuned L2 Stage 2
Scratch Baseline L0
Scratch Baseline L1
Scratch Baseline L2 Stage 2
Diversity 300K
Diversity 800K
−10⁰
0
10⁰
Drift Rate (meV/atom·ps) [symlog]
10²
10³
σ_T (K) [log scale]
10¹
10²
10³
Final MSD (Å²) [log scale]

Model Verdict & Architecture Selection

Best for Operational Stability & Efficiency: MH-1 LoRA

- Leverages universal multi-head priors.
- Completely bypasses the epoch-30 gradient shock.
- Maintains tight energy conservation and pristine radial distribution functions.

Best for Standard Domain Adaptation: MP-0 Fine-Tuned (L1 / L2 Stage 2)

- Successfully transfers Materials Project crystalline inorganic representations.
- Achieves strong structural rigidity and low potential energy drift.

Baseline Validation: Scratch Models

- Lacks pre-training head starts.
- Proves MACE architectures can successfully learn complex perovskite force fields entirely from ab initio targets.

Grazie



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